EXECUTIVE SUMMARY

> ***Quick Jump:*** *Use the links in the tables below to jump directly to the detailed images/reports in the rest of the document.*

1. Grid Search Configurations (Ranked)

Indian Pines (All Runs)

Rank	Patch Size	Batch Size	Overall Acc (OA)	Avg Acc (AA)	Kappa	Train Time	Details Link
1st	11x11	32	**98.05%**	94.82%	97.78	8.8 min	View Details
2nd	15x15	32	97.82%	94.68%	97.52	16.6 min	View Details
3rd	17x17	32	97.75%	91.40%	97.44	19.4 min	View Details
4th	13x13	32	97.63%	95.97%	97.29	12.0 min	View Details
5th	13x13	64	97.28%	97.17%	96.90	11.6 min	View Details
6th	15x15	64	97.26%	92.59%	96.88	16.4 min	View Details
7th	17x17	64	97.13%	90.56%	96.73	20.5 min	View Details
8th	11x11	64	97.07%	93.08%	96.66	11.4 min	View Details
9th	13x13	128	97.02%	96.69%	96.60	11.5 min	View Details
10th	15x15	128	96.98%	92.83%	96.56	16.3 min	View Details
11th	17x17	128	96.80%	90.93%	96.35	17.8 min	View Details
12th	15x15	256	96.18%	94.14%	95.64	14.6 min	View Details
13th	11x11	128	96.16%	91.54%	95.63	11.8 min	View Details
14th	9x9	64	96.12%	95.35%	95.57	7.8 min	View Details
15th	9x9	32	95.79%	92.75%	95.20	6.0 min	View Details
16th	17x17	256	95.52%	91.08%	94.89	19.1 min	View Details
17th	13x13	256	95.37%	95.61%	94.72	10.1 min	View Details
18th	9x9	128	94.95%	93.30%	94.24	8.0 min	View Details

Rank	Patch Size	Batch Size	Overall Acc (OA)	Avg Acc (AA)	Kappa	Train Time	Details Link
19th	11x11	256	94.62%	93.04%	93.87	12.5 min	View Details
20th	9x9	256	92.61%	93.10%	91.56	8.3 min	View Details

Pavia University (All Runs)

Rank	Patch Size	Batch Size	Overall Acc (OA)	Avg Acc (AA)	Kappa	Train Time	Details Link
1st	13x13	32	**99.86%**	99.71%	99.82	25.6 min	View Details
2nd	13x13	64	99.86%	99.70%	99.81	15.4 min	View Details
3rd	13x13	128	99.84%	99.70%	99.79	28.2 min	View Details
4th	15x15	32	99.84%	99.65%	99.78	38.0 min	View Details
5th	9x9	32	99.83%	99.82%	99.78	14.1 min	View Details
6th	11x11	32	99.82%	99.71%	99.76	12.9 min	View Details
7th	9x9	64	99.80%	99.74%	99.73	15.2 min	View Details
8th	13x13	256	99.78%	99.55%	99.70	24.4 min	View Details
9th	11x11	128	99.76%	99.70%	99.69	23.4 min	View Details
10th	11x11	64	99.75%	99.66%	99.68	19.7 min	View Details
11th	15x15	64	99.75%	99.59%	99.66	23.6 min	View Details
12th	15x15	128	99.73%	99.53%	99.64	29.6 min	View Details
13th	11x11	256	99.72%	99.52%	99.62	28.3 min	View Details
14th	15x15	256	99.72%	99.54%	99.62	44.4 min	View Details
15th	17x17	32	99.72%	99.31%	99.62	24.0 min	View Details
16th	17x17	128	99.71%	99.46%	99.62	41.2 min	View Details
17th	17x17	64	99.70%	99.50%	99.60	32.1 min	View Details
18th	17x17	256	99.63%	99.19%	99.52	38.2 min	View Details
19th	9x9	128	99.62%	99.66%	99.50	15.8 min	View Details
20th	9x9	256	99.42%	99.44%	99.24	16.1 min	View Details

Salinas Valley (All Runs)

Rank	Patch Size	Batch Size	Overall Acc (OA)	Avg Acc (AA)	Kappa	Train Time	Details Link
1st	17x17	32	**100.00%**	100.00%	100.00	116.8 min	View Details
2nd	17x17	64	100.00%	99.99%	100.00	115.4 min	View Details
3rd	15x15	64	99.98%	99.97%	99.98	93.9 min	View Details
4th	17x17	128	99.97%	99.94%	99.97	94.2 min	View Details
5th	17x17	256	99.97%	99.94%	99.97	111.0 min	View Details
6th	13x13	64	99.97%	99.96%	99.97	57.3 min	View Details
7th	15x15	32	99.97%	99.94%	99.96	63.5 min	View Details
8th	15x15	128	99.96%	99.93%	99.96	95.1 min	View Details
9th	13x13	128	99.95%	99.93%	99.94	67.0 min	View Details
10th	13x13	32	99.92%	99.92%	99.91	43.4 min	View Details
11th	11x11	32	99.92%	99.92%	99.91	57.7 min	View Details
12th	11x11	64	99.92%	99.92%	99.91	69.7 min	View Details
13th	15x15	256	99.92%	99.82%	99.91	74.6 min	View Details
14th	11x11	256	99.91%	99.93%	99.90	79.5 min	View Details
15th	13x13	256	99.88%	99.86%	99.86	59.5 min	View Details
16th	11x11	128	99.87%	99.89%	99.85	65.4 min	View Details
17th	9x9	128	99.83%	99.82%	99.81	49.7 min	View Details
18th	9x9	32	99.83%	99.87%	99.81	39.8 min	View Details
19th	9x9	64	99.76%	99.81%	99.74	40.4 min	View Details
20th	9x9	256	99.60%	99.73%	99.56	54.5 min	View Details

2. Train Ratio Scarcity Experiment (Side-by-Side)

(All 12 Scarcity tests were performed using a constant configuration of Patch Size: 11x11 and Batch Size: 32 to isolate the effect of data reduction).

Dataset	1% Ratio OA	2% Ratio OA	5% Ratio OA	10% Ratio OA	Accuracy Drop (10% to 1%)
Indian Pines	70.12%	81.55%	93.91%	**97.84%**	-27.73%

Dataset	1% Ratio OA	2% Ratio OA	5% Ratio OA	10% Ratio OA	Accuracy Drop (10% to 1%)
Pavia University	95.41%	98.04%	99.51%	**99.83%**	-4.43%
Salinas Valley	97.73%	99.11%	99.79%	**99.93%**	-2.20%

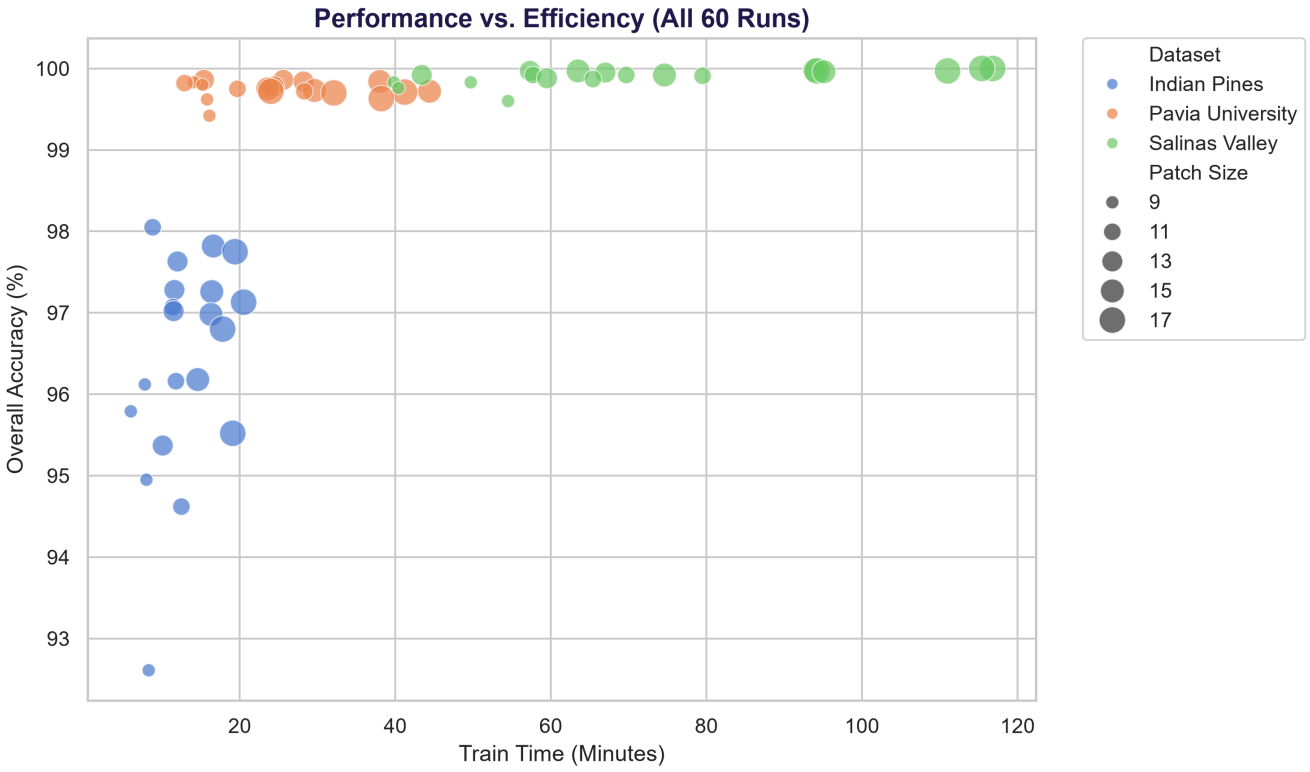
3. Class-Level Extremes (Under Extreme 1% Scarcity)

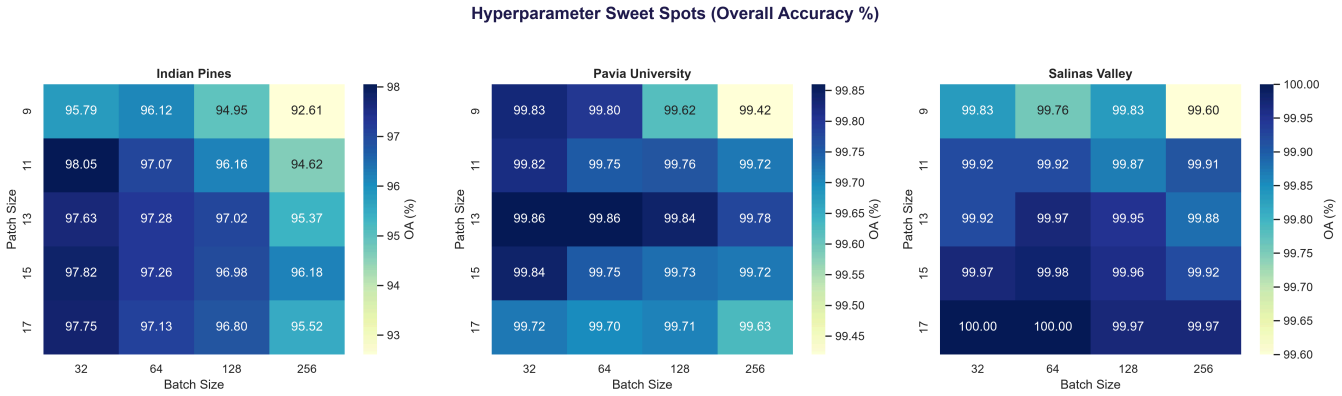
Dataset	Best Performing Class	Worst Performing Class
Indian Pines	Hay-windrowed (99.27%)	Oats (7.66%)
Pavia University	Painted metal sheets (99.62%)	Self-Blocking Bricks (84.70%)
Salinas Valley	Brocoli_green_weeds_2 (100.00%)	Fallow (89.47%)

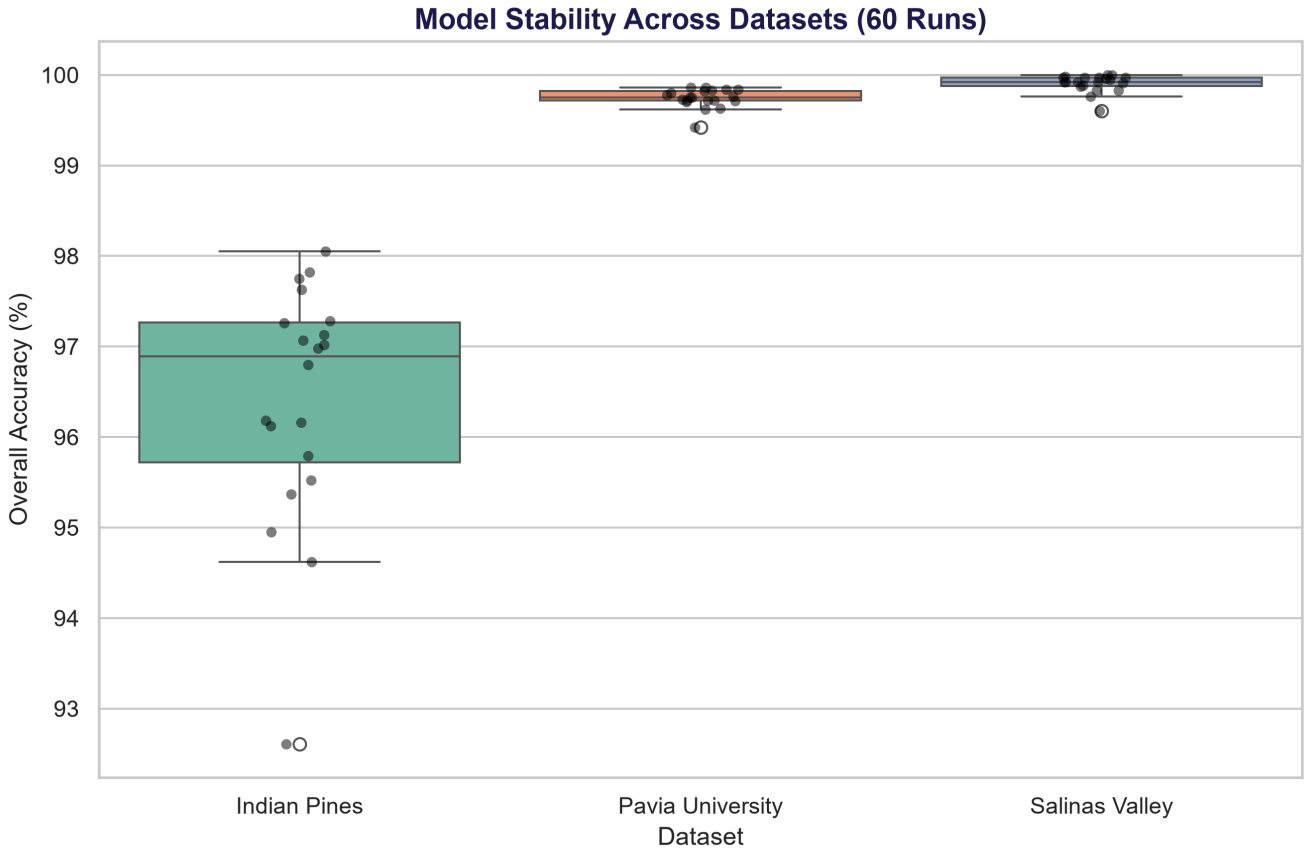
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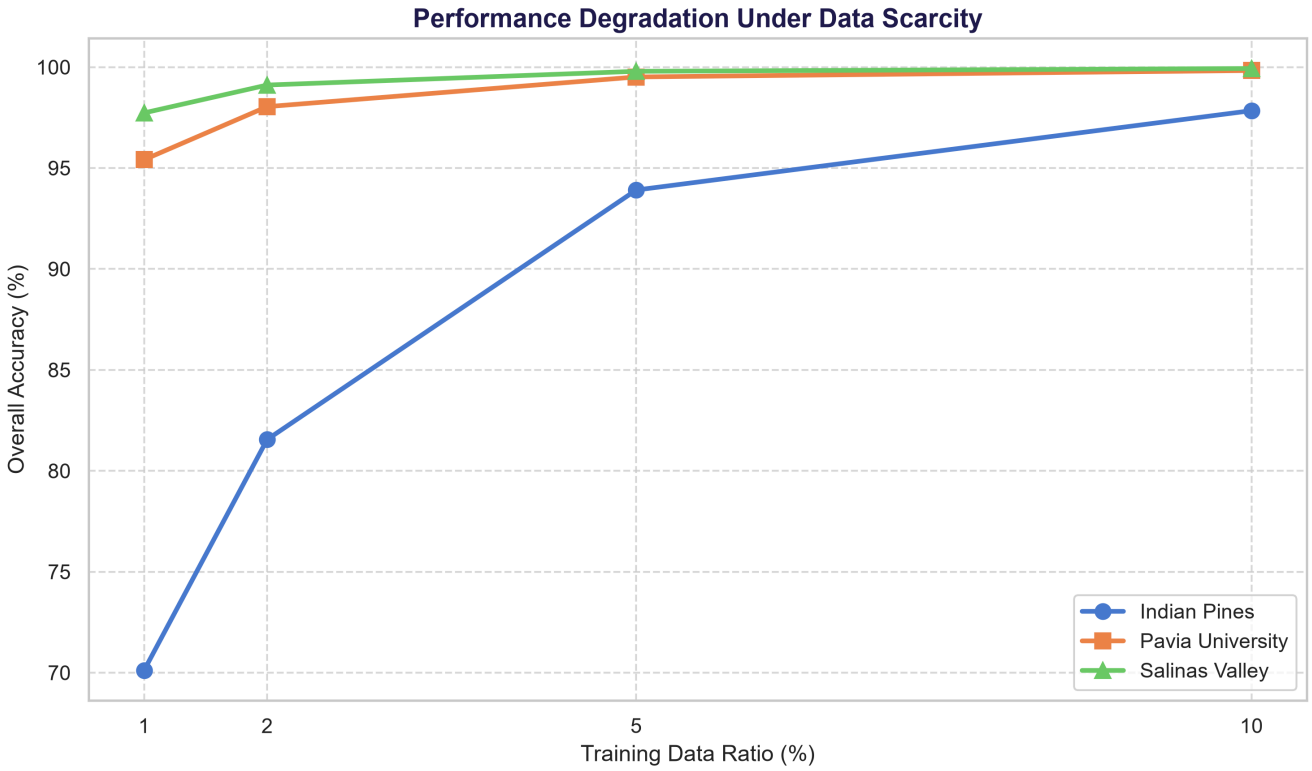
4. Visual Summary of All Runs

Below is the definitive visual summary of the hyperparameter sweeps (Patch Size & Batch Size) and the data scarcity tests (Train Ratio). These visualizations compress the 72 distinct runs into four key insights: performance efficiency, hyperparameter sweet spots, cross-dataset stability, and robustness under extreme data starvation.




Hyperparameter Sweet Spots (Overall Accuracy %)
Indian Pines
Pavia University
Salinas Valley
Patch Size
Batch Size
OA (%)
9
11
13
15
17
32
64
128
256
95.79 96.12 94.95 92.61
98.05 97.07 96.16 94.62
97.63 97.28 97.02 95.37
97.82 97.26 96.98 96.18
97.75 97.13 96.80 95.52
99.83 99.80 99.62 99.42
99.82 99.75 99.76 99.72
99.86 99.86 99.84 99.78
99.84 99.75 99.73 99.72
99.72 99.70 99.71 99.63
99.83 99.76 99.83 99.60
99.92 99.92 99.87 99.91
99.92 99.97 99.95 99.88
99.97 99.98 99.96 99.92
100.00 100.00 99.97 99.97


Model Stability Across Datasets (60 Runs)
Overall Accuracy (%)
100
99
98
97
96
95
94
93
Indian Pines
Pavia University
Salinas Valley
Dataset



PART 1: Full 60-Run Grid Search Results

Indian Pines

Patch Size 9x9 | Batch Size 32

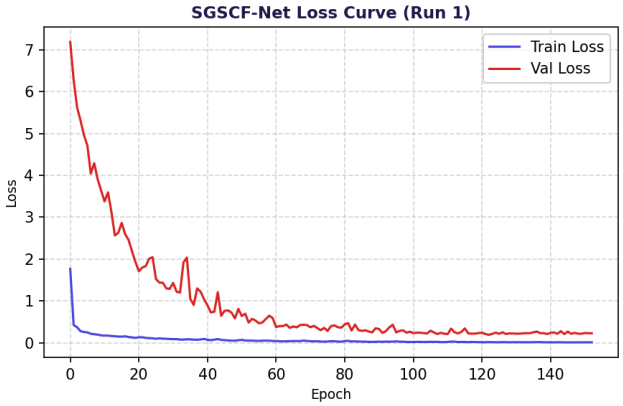
Overall Metrics:
- **OA (Overall Accuracy):** 95.79%
- **AA (Average Accuracy):** 92.75%
- **Kappa Coefficient:** 95.20
- **Train Time:** 6.0 min

Class-wise Accuracy:
- Alfalfa: 91.67%
- Corn-notill: 97.32%
- Corn-mintill: 97.97%
- Corn: 97.25%
- Grass-pasture: 96.43%
- Grass-trees: 98.83%
- Grass-pasture-mowed: 75.00%
- Hay-windrowed: 98.19%
- Oats: 73.68%
- Soybean-notill: 90.85%
- Soybean-mintill: 95.18%
- Soybean-clean: 89.53%

- Wheat: 97.60%
- Woods: 98.45%
- Buildings-Grass-Trees: 97.36%
- Stone-Steel-Towers: 88.75%

Visualizations:



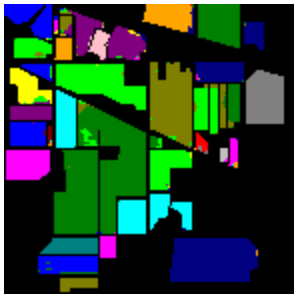


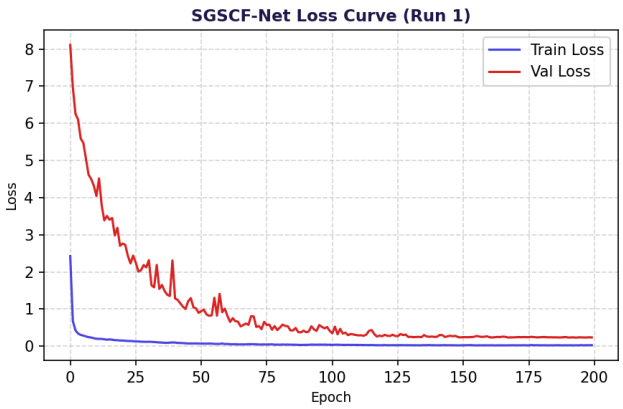
Patch Size 9x9 | Batch Size 64

Overall Metrics:
- **OA (Overall Accuracy):** 96.12%
- **AA (Average Accuracy):** 95.35%
- **Kappa Coefficient:** 95.57
- **Train Time:** 7.8 min

Class-wise Accuracy:
- Alfalfa: 96.77%
- Corn-notill: 95.48%
- Corn-mintill: 98.74%
- Corn: 90.71%
- Grass-pasture: 97.68%
- Grass-trees: 99.66%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 97.94%
- Oats: 81.25%
- Soybean-notill: 92.54%
- Soybean-mintill: 95.94%
- Soybean-clean: 94.32%
- Wheat: 100.00%
- Woods: 96.76%
- Buildings-Grass-Trees: 95.39%
- Stone-Steel-Towers: 92.41%

Visualizations:





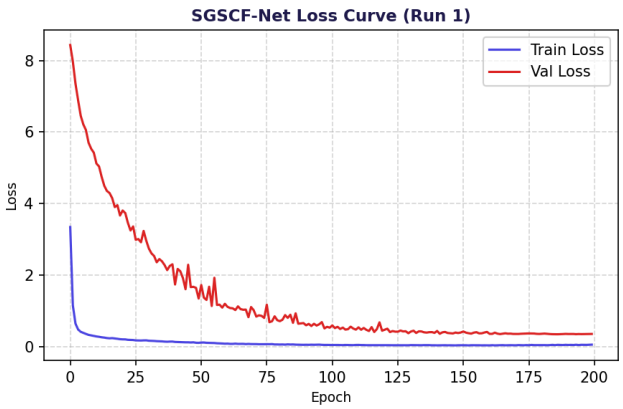
Patch Size 9x9 | Batch Size 128

Overall Metrics:
- **OA (Overall Accuracy):** 94.95%
- **AA (Average Accuracy):** 93.30%
- **Kappa Coefficient:** 94.24
- **Train Time:** 8.0 min

Class-wise Accuracy:
- Alfalfa: 89.19%
- Corn-notill: 92.74%
- Corn-mintill: 96.83%
- Corn: 94.77%
- Grass-pasture: 97.95%
- Grass-trees: 99.50%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 100.00%
- Oats: 60.00%
- Soybean-notill: 93.12%
- Soybean-mintill: 94.23%
- Soybean-clean: 86.00%
- Wheat: 99.39%
- Woods: 97.04%
- Buildings-Grass-Trees: 97.32%
- Stone-Steel-Towers: 94.67%

Visualizations:





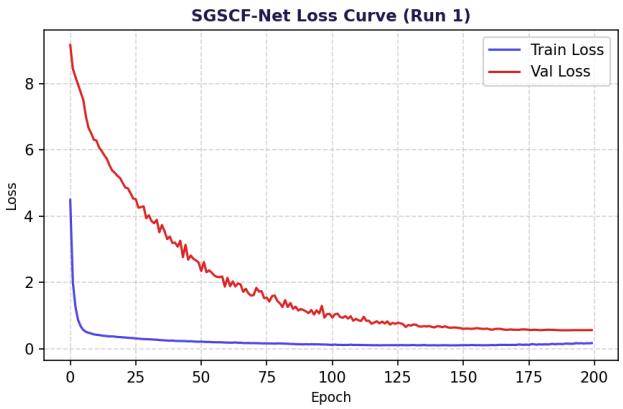
Patch Size 9x9 | Batch Size 256

Overall Metrics:
- **OA (Overall Accuracy):** 92.61%
- **AA (Average Accuracy):** 93.10%
- **Kappa Coefficient:** 91.56
- **Train Time:** 8.3 min

Class-wise Accuracy:
- Alfalfa: 100.00%
- Corn-notill: 87.36%
- Corn-mintill: 95.91%
- Corn: 92.42%
- Grass-pasture: 95.84%
- Grass-trees: 98.49%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 99.74%
- Oats: 72.73%
- Soybean-notill: 87.77%
- Soybean-mintill: 92.63%
- Soybean-clean: 82.03%
- Wheat: 96.88%
- Woods: 95.83%
- Buildings-Grass-Trees: 96.32%
- Stone-Steel-Towers: 95.71%

Visualizations:





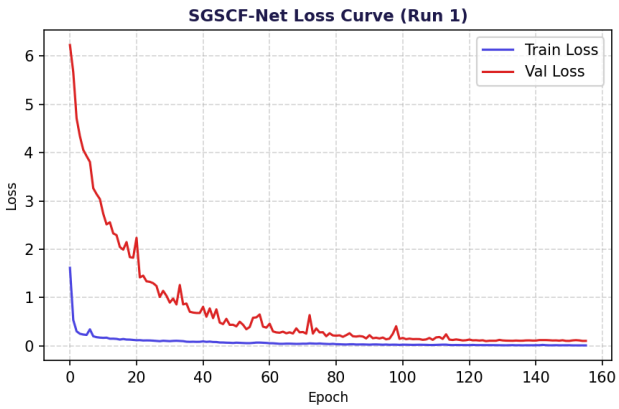
Patch Size 11x11 | Batch Size 32

Overall Metrics:
- **OA (Overall Accuracy):** 98.05%
- **AA (Average Accuracy):** 94.82%
- **Kappa Coefficient:** 97.78
- **Train Time:** 8.8 min

Class-wise Accuracy:
- Alfalfa: 95.00%
- Corn-notill: 98.13%
- Corn-mintill: 97.72%
- Corn: 100.00%
- Grass-pasture: 99.74%
- Grass-trees: 99.17%
- Grass-pasture-mowed: 95.24%
- Hay-windrowed: 100.00%
- Oats: 54.17%
- Soybean-notill: 98.84%
- Soybean-mintill: 98.42%
- Soybean-clean: 91.85%
- Wheat: 99.39%
- Woods: 98.65%
- Buildings-Grass-Trees: 99.00%
- Stone-Steel-Towers: 91.89%

Visualizations:





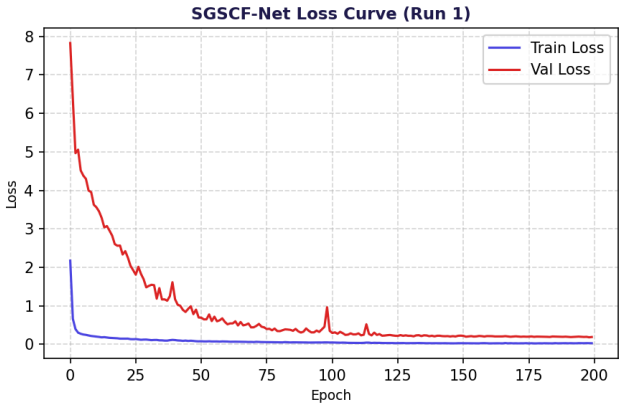
Patch Size 11x11 | Batch Size 64

Overall Metrics:
- **OA (Overall Accuracy):** 97.07%
- **AA (Average Accuracy):** 93.08%
- **Kappa Coefficient:** 96.66
- **Train Time:** 11.4 min

Class-wise Accuracy:
- Alfalfa: 92.86%
- Corn-notill: 98.06%
- Corn-mintill: 97.70%
- Corn: 97.42%
- Grass-pasture: 98.70%
- Grass-trees: 96.90%
- Grass-pasture-mowed: 69.23%
- Hay-windrowed: 100.00%
- Oats: 69.23%
- Soybean-notill: 92.27%
- Soybean-mintill: 98.01%
- Soybean-clean: 95.62%
- Wheat: 100.00%
- Woods: 98.18%
- Buildings-Grass-Trees: 96.71%
- Stone-Steel-Towers: 88.46%

Visualizations:





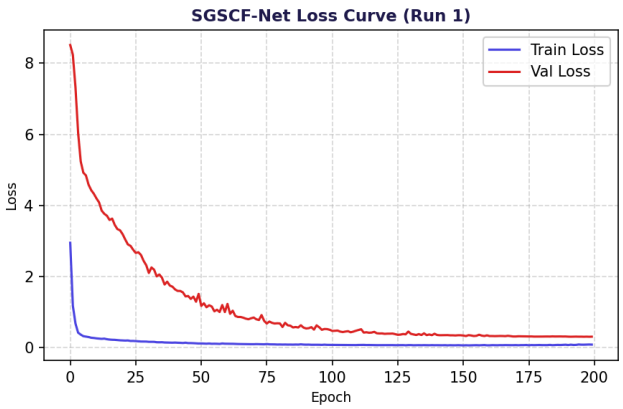
Patch Size 11x11 | Batch Size 128

Overall Metrics:
- **OA (Overall Accuracy):** 96.16%
- **AA (Average Accuracy):** 91.54%
- **Kappa Coefficient:** 95.63
- **Train Time:** 11.8 min

Class-wise Accuracy:
- Alfalfa: 100.00%
- Corn-notill: 95.83%
- Corn-mintill: 96.32%
- Corn: 95.77%
- Grass-pasture: 96.05%
- Grass-trees: 97.05%
- Grass-pasture-mowed: 72.00%
- Hay-windrowed: 98.42%
- Oats: 50.00%
- Soybean-notill: 92.59%
- Soybean-mintill: 97.59%
- Soybean-clean: 92.21%
- Wheat: 99.39%
- Woods: 97.98%
- Buildings-Grass-Trees: 97.67%
- Stone-Steel-Towers: 85.71%

Visualizations:





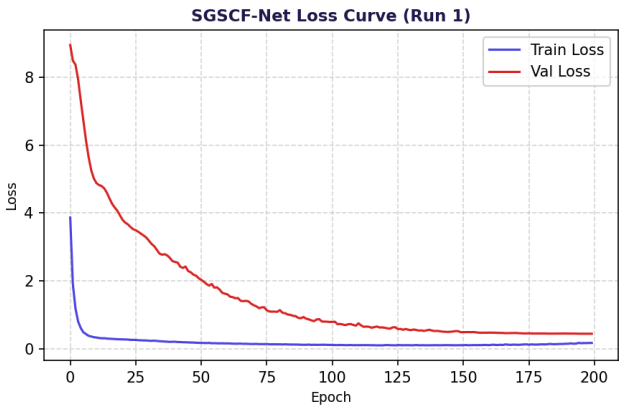
Patch Size 11x11 | Batch Size 256

Overall Metrics:
- **OA (Overall Accuracy):** 94.62%
- **AA (Average Accuracy):** 93.04%
- **Kappa Coefficient:** 93.87
- **Train Time:** 12.5 min

Class-wise Accuracy:
- Alfalfa: 100.00%
- Corn-notill: 95.43%
- Corn-mintill: 93.63%
- Corn: 85.92%
- Grass-pasture: 93.55%
- Grass-trees: 97.64%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 95.96%
- Oats: 71.43%
- Soybean-notill: 87.83%
- Soybean-mintill: 96.12%
- Soybean-clean: 90.53%
- Wheat: 100.00%
- Woods: 97.60%
- Buildings-Grass-Trees: 99.66%
- Stone-Steel-Towers: 83.33%

Visualizations:





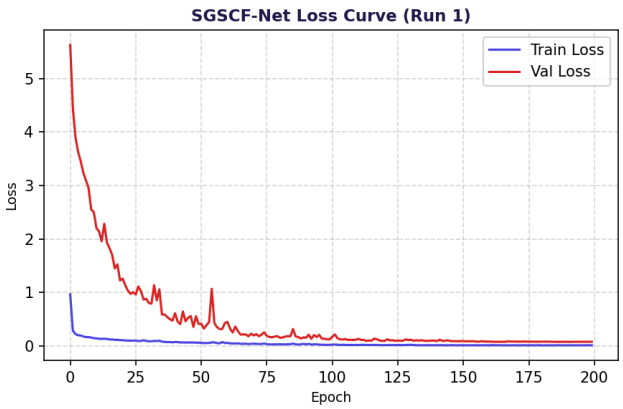
Patch Size 13x13 | Batch Size 32

Overall Metrics:
- **OA (Overall Accuracy):** 97.63%
- **AA (Average Accuracy):** 95.97%
- **Kappa Coefficient:** 97.29
- **Train Time:** 12.0 min

Class-wise Accuracy:
- Alfalfa: 90.70%
- Corn-notill: 98.27%
- Corn-mintill: 99.22%
- Corn: 100.00%
- Grass-pasture: 98.47%
- Grass-trees: 99.32%
- Grass-pasture-mowed: 95.24%
- Hay-windrowed: 100.00%
- Oats: 84.21%
- Soybean-notill: 94.35%
- Soybean-mintill: 96.48%
- Soybean-clean: 96.77%
- Wheat: 100.00%
- Woods: 99.90%
- Buildings-Grass-Trees: 97.18%
- Stone-Steel-Towers: 85.37%

Visualizations:





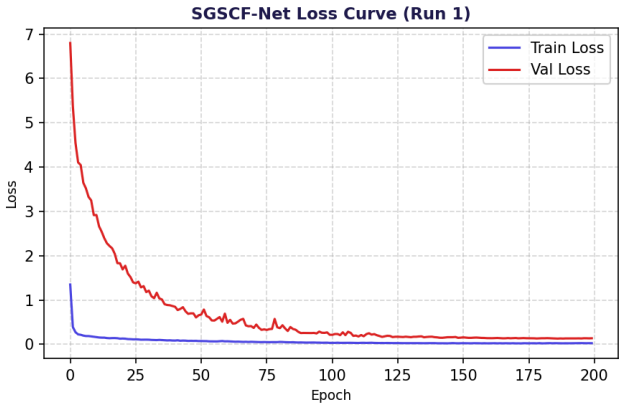
Patch Size 13x13 | Batch Size 64

Overall Metrics:
- **OA (Overall Accuracy):** 97.28%
- **AA (Average Accuracy):** 97.17%
- **Kappa Coefficient:** 96.90
- **Train Time:** 11.6 min

Class-wise Accuracy:
- Alfalfa: 95.00%
- Corn-notill: 98.99%
- Corn-mintill: 98.32%
- Corn: 100.00%
- Grass-pasture: 96.73%
- Grass-trees: 97.35%
- Grass-pasture-mowed: 95.24%
- Hay-windrowed: 100.00%
- Oats: 100.00%
- Soybean-notill: 96.52%
- Soybean-mintill: 95.62%
- Soybean-clean: 94.73%
- Wheat: 99.38%
- Woods: 99.22%
- Buildings-Grass-Trees: 95.67%
- Stone-Steel-Towers: 92.00%

Visualizations:





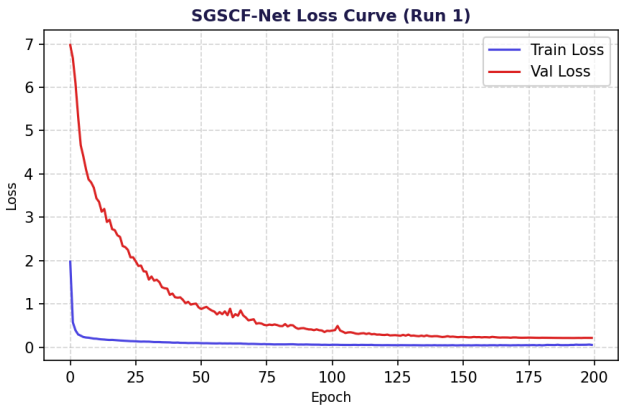
Patch Size 13x13 | Batch Size 128

Overall Metrics:
- **OA (Overall Accuracy):** 97.02%
- **AA (Average Accuracy):** 96.69%
- **Kappa Coefficient:** 96.60
- **Train Time:** 11.5 min

Class-wise Accuracy:
- Alfalfa: 97.44%
- Corn-notill: 98.33%
- Corn-mintill: 97.55%
- Corn: 100.00%
- Grass-pasture: 96.67%
- Grass-trees: 96.85%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 100.00%
- Oats: 90.91%
- Soybean-notill: 97.34%
- Soybean-mintill: 96.11%
- Soybean-clean: 92.80%
- Wheat: 99.38%
- Woods: 98.55%
- Buildings-Grass-Trees: 93.29%
- Stone-Steel-Towers: 91.78%

Visualizations:





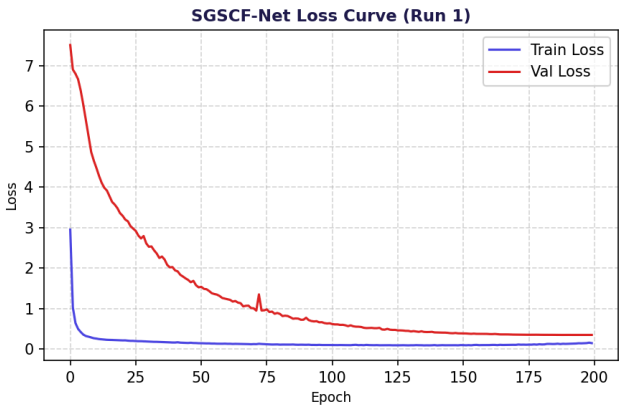
Patch Size 13x13 | Batch Size 256

Overall Metrics:
- **OA (Overall Accuracy):** 95.37%
- **AA (Average Accuracy):** 95.61%
- **Kappa Coefficient:** 94.72
- **Train Time:** 10.1 min

Class-wise Accuracy:
- Alfalfa: 100.00%
- Corn-notill: 96.18%
- Corn-mintill: 94.26%
- Corn: 99.42%
- Grass-pasture: 95.60%
- Grass-trees: 96.37%
- Grass-pasture-mowed: 87.50%
- Hay-windrowed: 100.00%
- Oats: 100.00%
- Soybean-notill: 94.88%
- Soybean-mintill: 94.11%
- Soybean-clean: 89.60%
- Wheat: 100.00%
- Woods: 98.65%
- Buildings-Grass-Trees: 90.61%
- Stone-Steel-Towers: 92.54%

Visualizations:





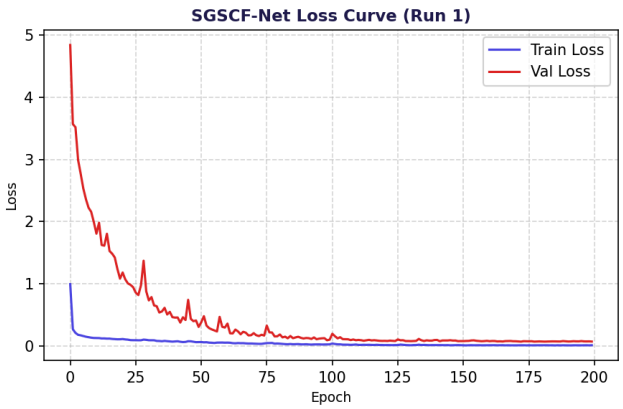
Patch Size 15x15 | Batch Size 32

Overall Metrics:
- **OA (Overall Accuracy):** 97.82%
- **AA (Average Accuracy):** 94.68%
- **Kappa Coefficient:** 97.52
- **Train Time:** 16.6 min

Class-wise Accuracy:
- Alfalfa: 76.09%
- Corn-notill: 98.36%
- Corn-mintill: 99.24%
- Corn: 100.00%
- Grass-pasture: 99.23%
- Grass-trees: 95.95%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 100.00%
- Oats: 71.43%
- Soybean-notill: 95.58%
- Soybean-mintill: 98.57%
- Soybean-clean: 93.83%
- Wheat: 100.00%
- Woods: 99.71%
- Buildings-Grass-Trees: 97.17%
- Stone-Steel-Towers: 89.74%

Visualizations:





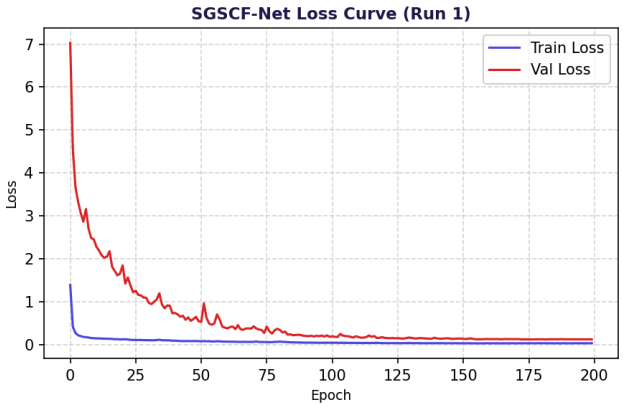
Patch Size 15x15 | Batch Size 64

Overall Metrics:
- **OA (Overall Accuracy):** 97.26%
- **AA (Average Accuracy):** 92.59%
- **Kappa Coefficient:** 96.88
- **Train Time:** 16.4 min

Class-wise Accuracy:
- Alfalfa: 64.15%
- Corn-notill: 98.62%
- Corn-mintill: 98.49%
- Corn: 100.00%
- Grass-pasture: 98.97%
- Grass-trees: 95.66%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 100.00%
- Oats: 58.33%
- Soybean-notill: 95.98%
- Soybean-mintill: 97.90%
- Soybean-clean: 93.16%
- Wheat: 100.00%
- Woods: 99.22%
- Buildings-Grass-Trees: 93.37%
- Stone-Steel-Towers: 87.50%

Visualizations:





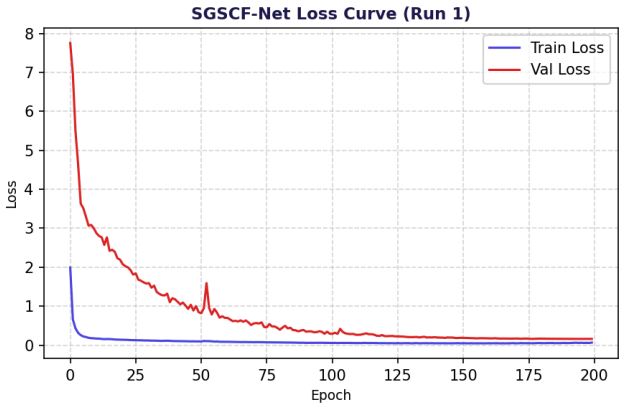
Patch Size 15x15 | Batch Size 128

Overall Metrics:
- **OA (Overall Accuracy):** 96.98%
- **AA (Average Accuracy):** 92.83%
- **Kappa Coefficient:** 96.56
- **Train Time:** 16.3 min

Class-wise Accuracy:
- Alfalfa: 82.61%
- Corn-notill: 98.16%
- Corn-mintill: 98.18%
- Corn: 100.00%
- Grass-pasture: 97.45%
- Grass-trees: 97.85%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 100.00%
- Oats: 54.55%
- Soybean-notill: 97.18%
- Soybean-mintill: 96.87%
- Soybean-clean: 90.87%
- Wheat: 100.00%
- Woods: 98.84%
- Buildings-Grass-Trees: 93.11%
- Stone-Steel-Towers: 79.55%

Visualizations:





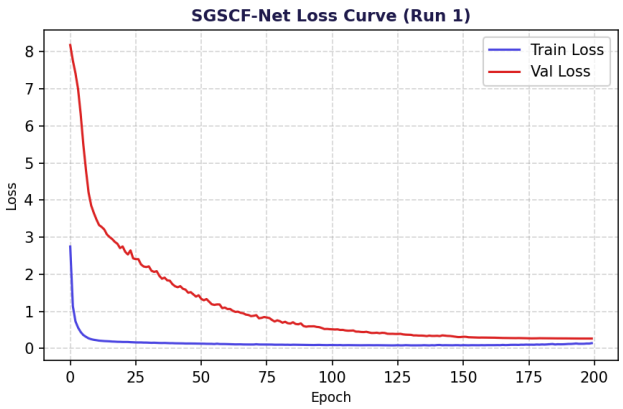
Patch Size 15x15 | Batch Size 256

Overall Metrics:
- **OA (Overall Accuracy):** 96.18%
- **AA (Average Accuracy):** 94.14%
- **Kappa Coefficient:** 95.64
- **Train Time:** 14.6 min

Class-wise Accuracy:
- Alfalfa: 100.00%
- Corn-notill: 98.68%
- Corn-mintill: 96.95%
- Corn: 96.83%
- Grass-pasture: 95.26%
- Grass-trees: 93.00%
- Grass-pasture-mowed: 100.00%
- Hay-windrowed: 99.73%
- Oats: 66.67%
- Soybean-notill: 96.59%
- Soybean-mintill: 95.49%
- Soybean-clean: 93.57%
- Wheat: 100.00%
- Woods: 98.17%
- Buildings-Grass-Trees: 91.74%
- Stone-Steel-Towers: 83.54%

Visualizations:





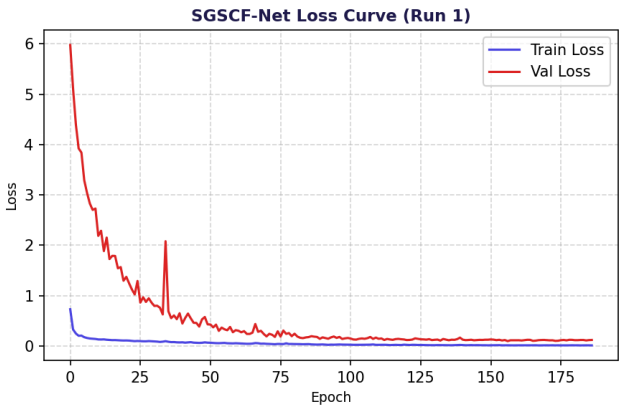
Patch Size 17x17 | Batch Size 32

Overall Metrics:
- **OA (Overall Accuracy):** 97.75%
- **AA (Average Accuracy):** 91.40%
- **Kappa Coefficient:** 97.44
- **Train Time:** 19.4 min

Class-wise Accuracy:
- Alfalfa: 72.00%
- Corn-notill: 97.93%
- Corn-mintill: 98.34%
- Corn: 100.00%
- Grass-pasture: 97.48%
- Grass-trees: 98.32%
- Grass-pasture-mowed: 84.21%
- Hay-windrowed: 99.74%
- Oats: 47.37%
- Soybean-notill: 97.18%
- Soybean-mintill: 98.31%
- Soybean-clean: 97.37%
- Wheat: 100.00%
- Woods: 99.80%
- Buildings-Grass-Trees: 95.69%
- Stone-Steel-Towers: 78.72%

Visualizations:





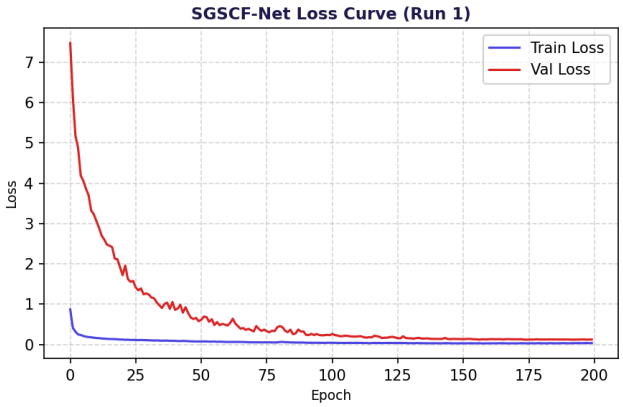
Patch Size 17x17 | Batch Size 64

Overall Metrics:
- **OA (Overall Accuracy):** 97.13%
- **AA (Average Accuracy):** 90.56%
- **Kappa Coefficient:** 96.73
- **Train Time:** 20.5 min

Class-wise Accuracy:
- Alfalfa: 73.91%
- Corn-notill: 98.33%
- Corn-mintill: 98.18%
- Corn: 100.00%
- Grass-pasture: 98.98%
- Grass-trees: 98.49%
- Grass-pasture-mowed: 79.17%
- Hay-windrowed: 99.73%
- Oats: 43.75%
- Soybean-notill: 95.08%
- Soybean-mintill: 97.22%
- Soybean-clean: 93.22%
- Wheat: 100.00%
- Woods: 99.42%
- Buildings-Grass-Trees: 95.38%
- Stone-Steel-Towers: 78.16%

Visualizations:





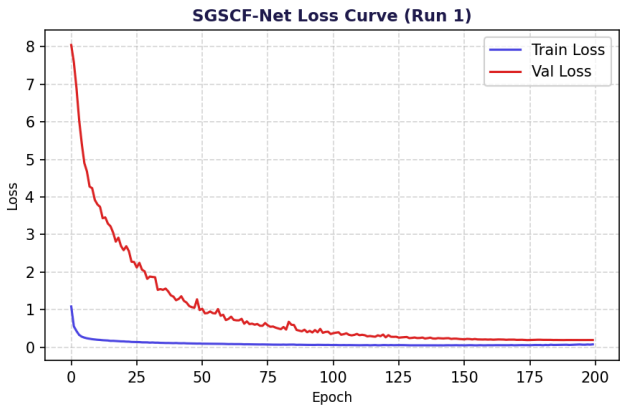
Patch Size 17x17 | Batch Size 128

Overall Metrics:
- **OA (Overall Accuracy):** 96.80%
- **AA (Average Accuracy):** 90.93%
- **Kappa Coefficient:** 96.35
- **Train Time:** 17.8 min

Class-wise Accuracy:
- Alfalfa: 91.43%
- Corn-notill: 98.66%
- Corn-mintill: 96.87%
- Corn: 100.00%
- Grass-pasture: 98.23%
- Grass-trees: 98.14%
- Grass-pasture-mowed: 68.97%
- Hay-windrowed: 100.00%
- Oats: 44.44%
- Soybean-notill: 93.25%
- Soybean-mintill: 96.53%
- Soybean-clean: 94.58%
- Wheat: 100.00%
- Woods: 98.07%
- Buildings-Grass-Trees: 99.04%
- Stone-Steel-Towers: 76.74%

Visualizations:





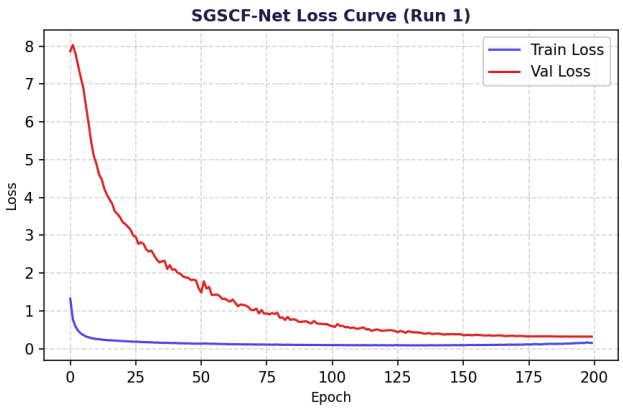
Patch Size 17x17 | Batch Size 256

Overall Metrics:
- **OA (Overall Accuracy):** 95.52%
- **AA (Average Accuracy):** 91.08%
- **Kappa Coefficient:** 94.89
- **Train Time:** 19.1 min

Class-wise Accuracy:
- Alfalfa: 91.67%
- Corn-notill: 99.21%
- Corn-mintill: 92.55%
- Corn: 100.00%
- Grass-pasture: 96.18%
- Grass-trees: 96.79%
- Grass-pasture-mowed: 76.92%
- Hay-windrowed: 100.00%
- Oats: 50.00%
- Soybean-notill: 92.62%
- Soybean-mintill: 94.46%
- Soybean-clean: 92.54%
- Wheat: 99.36%
- Woods: 96.96%
- Buildings-Grass-Trees: 98.06%
- Stone-Steel-Towers: 80.00%

Visualizations:





Pavia University

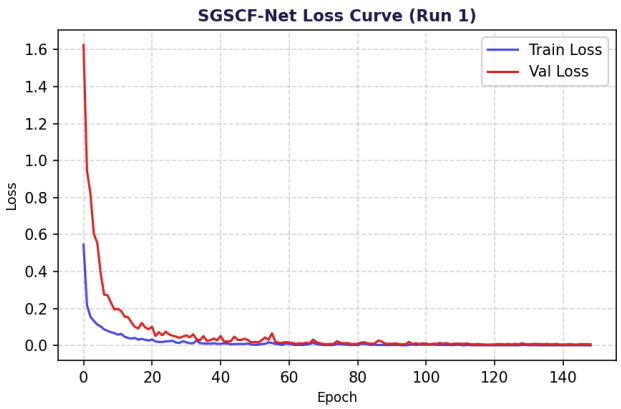
Patch Size 9x9 | Batch Size 32

Overall Metrics:
- **OA (Overall Accuracy):** 99.83%
- **AA (Average Accuracy):** 99.82%
- **Kappa Coefficient:** 99.78
- **Train Time:** 14.1 min

Class-wise Accuracy:
- Asphalt: 99.81%
- Meadows: 99.93%
- Gravel: 99.88%
- Trees: 99.31%
- Painted metal sheets: 100.00%
- Bare Soil: 99.83%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.62%
- Shadows: 100.00%

Visualizations:





Patch Size 9x9 | Batch Size 64

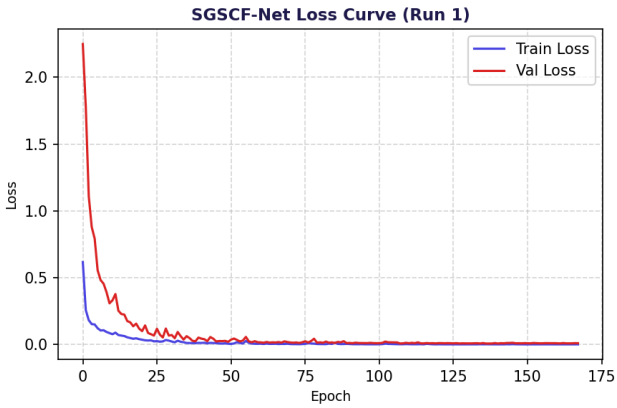
Overall Metrics:
- **OA (Overall Accuracy):** 99.80%
- **AA (Average Accuracy):** 99.74%
- **Kappa Coefficient:** 99.73
- **Train Time:** 15.2 min

Class-wise Accuracy:
- Asphalt: 99.89%
- Meadows: 99.97%
- Gravel: 99.70%
- Trees: 99.31%
- Painted metal sheets: 100.00%
- Bare Soil: 99.80%

- Bitumen: 100.00%
- Self-Blocking Bricks: 99.08%
- Shadows: 99.87%

Visualizations:





Patch Size 9x9 | Batch Size 128

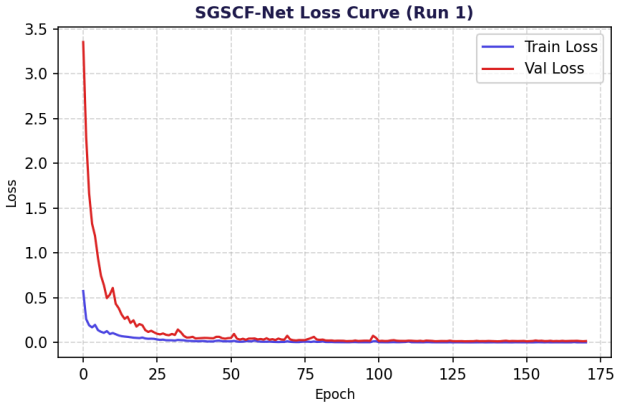
Overall Metrics:
- **OA (Overall Accuracy):** 99.62%
- **AA (Average Accuracy):** 99.66%
- **Kappa Coefficient:** 99.50
- **Train Time:** 15.8 min

Class-wise Accuracy:
- Asphalt: 99.39%
- Meadows: 99.76%

- Gravel: 100.00%
- Trees: 99.47%
- Painted metal sheets: 99.91%
- Bare Soil: 99.52%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.12%
- Shadows: 99.73%

Visualizations:



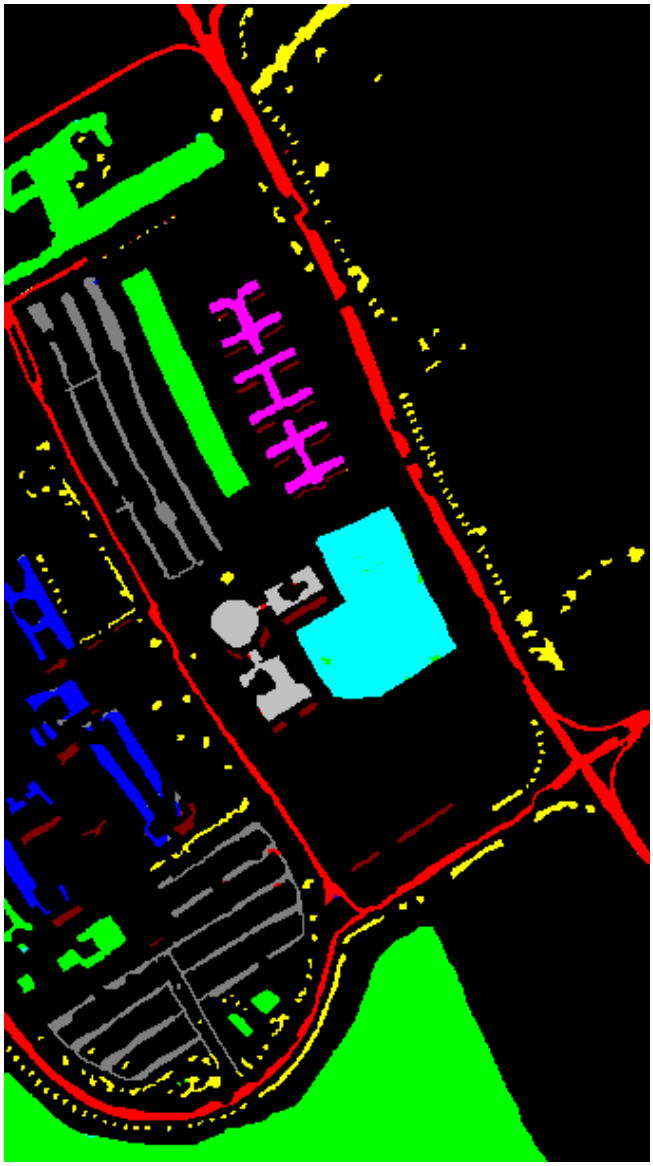


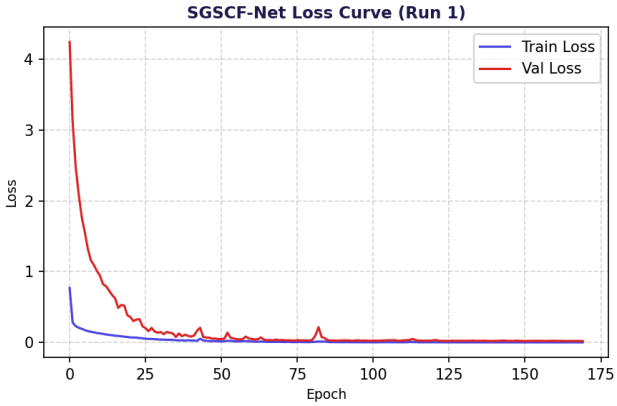
Patch Size 9x9 | Batch Size 256

Overall Metrics:
- **OA (Overall Accuracy):** 99.42%
- **AA (Average Accuracy):** 99.44%
- **Kappa Coefficient:** 99.24
- **Train Time:** 16.1 min

Class-wise Accuracy:
- Asphalt: 99.20%
- Meadows: 99.66%
- Gravel: 99.63%
- Trees: 99.31%
- Painted metal sheets: 100.00%
- Bare Soil: 99.60%
- Bitumen: 99.71%
- Self-Blocking Bricks: 97.97%
- Shadows: 99.87%

Visualizations:





Patch Size 11x11 | Batch Size 32

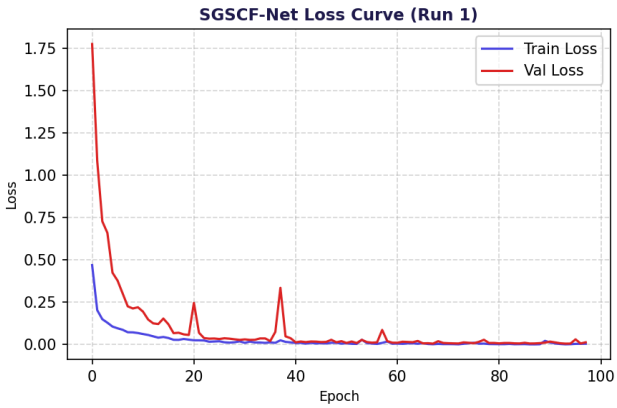
Overall Metrics:
- **OA (Overall Accuracy):** 99.82%

- **AA (Average Accuracy):** 99.71%
- **Kappa Coefficient:** 99.76
- **Train Time:** 12.9 min

Class-wise Accuracy:
- Asphalt: 99.79%
- Meadows: 100.00%
- Gravel: 99.76%
- Trees: 99.39%
- Painted metal sheets: 100.00%
- Bare Soil: 99.80%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.32%
- Shadows: 99.34%

Visualizations:



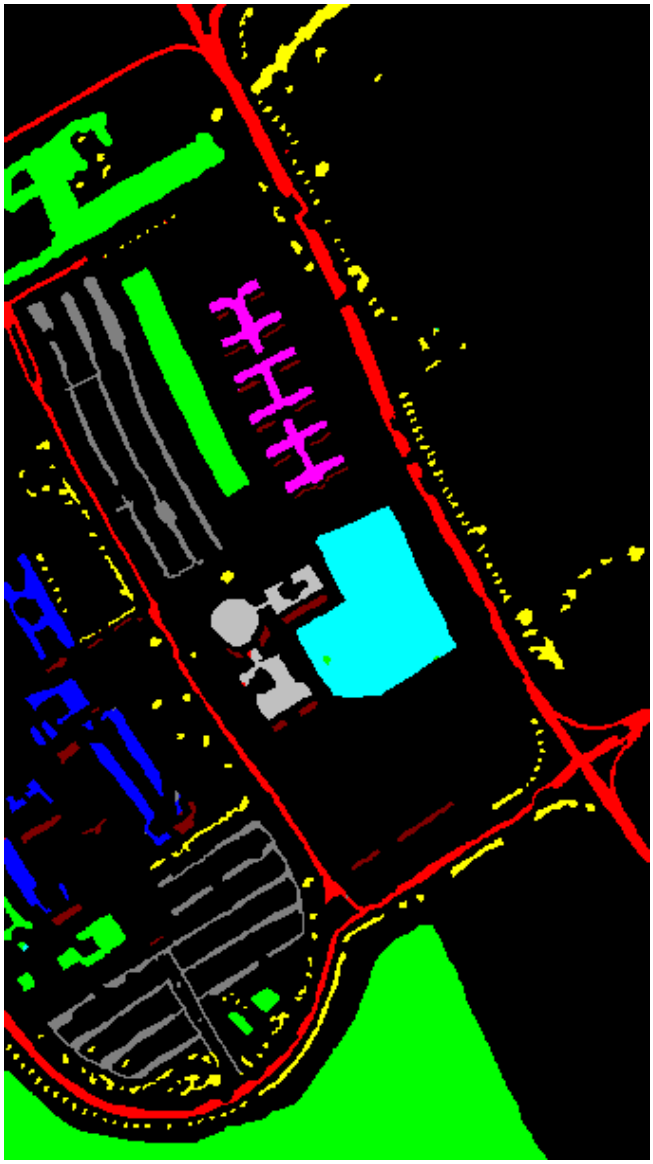


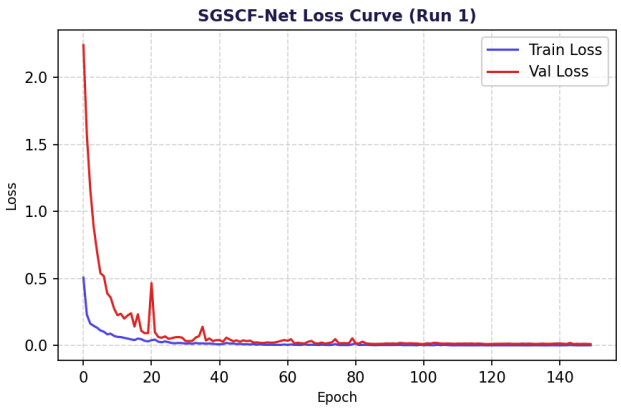
Patch Size 11x11 | Batch Size 64

Overall Metrics:
- **OA (Overall Accuracy):** 99.75%
- **AA (Average Accuracy):** 99.66%
- **Kappa Coefficient:** 99.68
- **Train Time:** 19.7 min

Class-wise Accuracy:
- Asphalt: 99.72%
- Meadows: 99.88%
- Gravel: 100.00%
- Trees: 99.47%
- Painted metal sheets: 99.91%
- Bare Soil: 99.85%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.25%
- Shadows: 98.82%

Visualizations:





Patch Size 11x11 | Batch Size 128

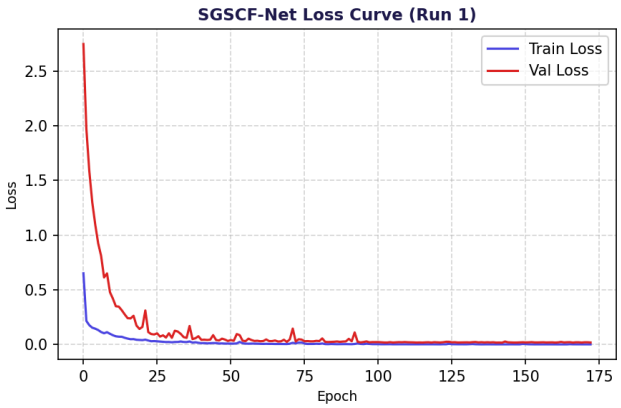
Overall Metrics:
- **OA (Overall Accuracy):** 99.76%
- **AA (Average Accuracy):** 99.70%
- **Kappa Coefficient:** 99.69
- **Train Time:** 23.4 min

Class-wise Accuracy:
- Asphalt: 99.59%
- Meadows: 100.00%
- Gravel: 99.76%
- Trees: 99.11%
- Painted metal sheets: 99.91%
- Bare Soil: 99.80%

- Bitumen: 100.00%
- Self-Blocking Bricks: 99.22%
- Shadows: 99.87%

Visualizations:





Patch Size 11x11 | Batch Size 256

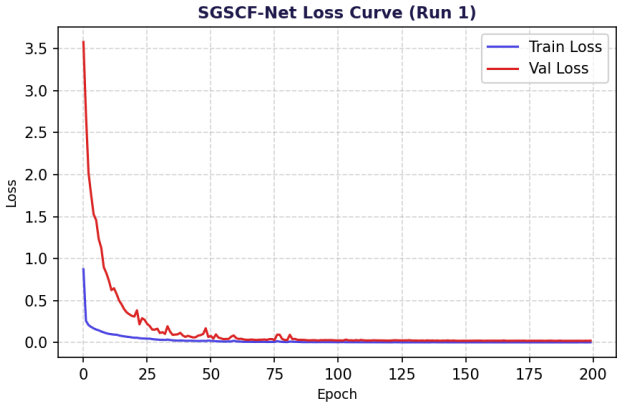
Overall Metrics:
- **OA (Overall Accuracy):** 99.72%
- **AA (Average Accuracy):** 99.52%
- **Kappa Coefficient:** 99.62
- **Train Time:** 28.3 min

Class-wise Accuracy:
- Asphalt: 99.72%
- Meadows: 99.97%

- Gravel: 99.88%
- Trees: 99.19%
- Painted metal sheets: 99.91%
- Bare Soil: 99.60%
- Bitumen: 99.81%
- Self-Blocking Bricks: 99.18%
- Shadows: 98.42%

Visualizations:





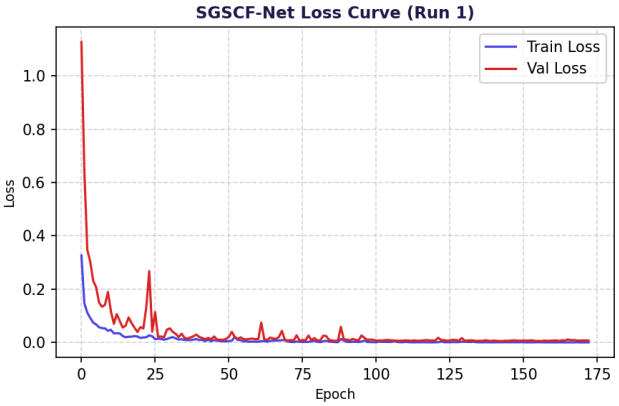
Patch Size 13x13 | Batch Size 32

Overall Metrics:
- **OA (Overall Accuracy):** 99.86%
- **AA (Average Accuracy):** 99.71%
- **Kappa Coefficient:** 99.82
- **Train Time:** 25.6 min

Class-wise Accuracy:
- Asphalt: 99.98%
- Meadows: 100.00%
- Gravel: 100.00%
- Trees: 99.23%
- Painted metal sheets: 100.00%
- Bare Soil: 99.85%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.62%
- Shadows: 98.69%

Visualizations:





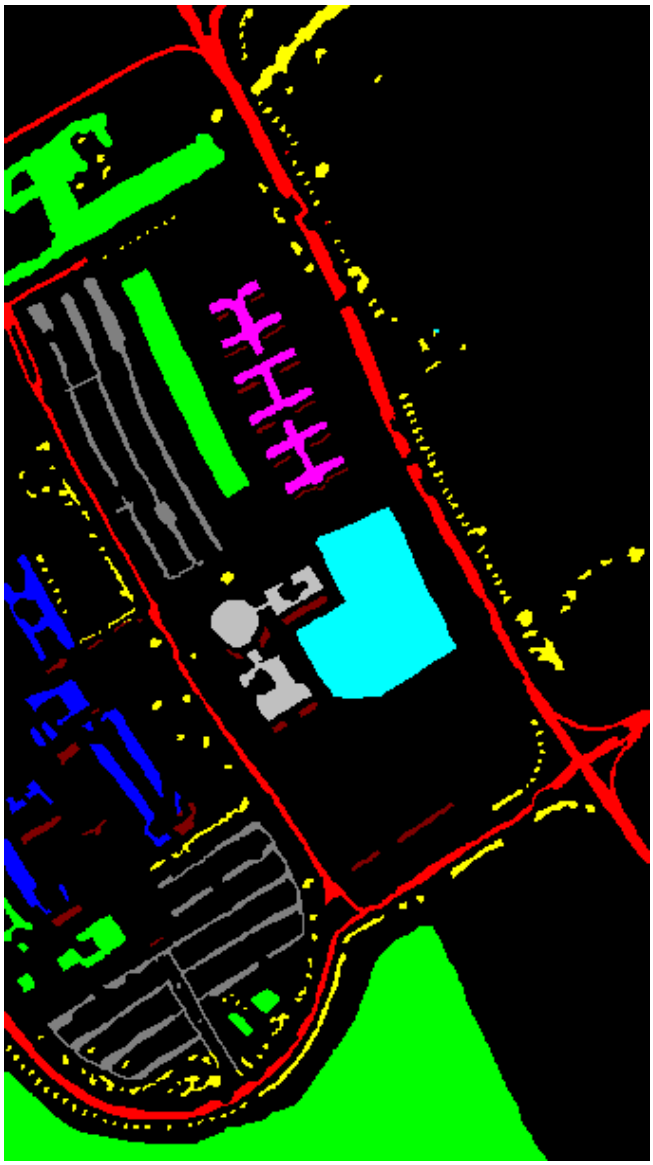
Patch Size 13x13 | Batch Size 64

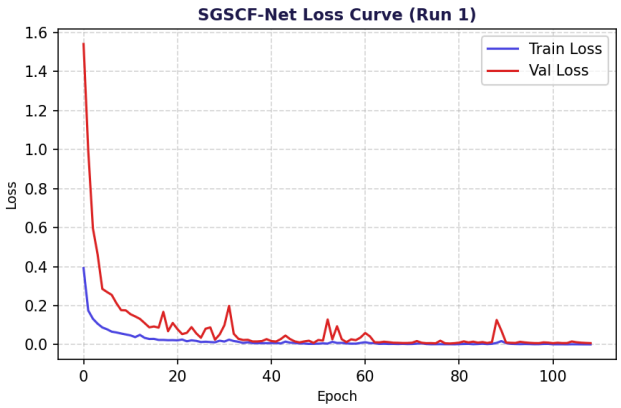
Overall Metrics:
- **OA (Overall Accuracy):** 99.86%

- **AA (Average Accuracy):** 99.70%
- **Kappa Coefficient:** 99.81
- **Train Time:** 15.4 min

Class-wise Accuracy:
- Asphalt: 99.91%
- Meadows: 100.00%
- Gravel: 100.00%
- Trees: 99.23%
- Painted metal sheets: 100.00%
- Bare Soil: 99.90%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.66%
- Shadows: 98.56%

Visualizations:



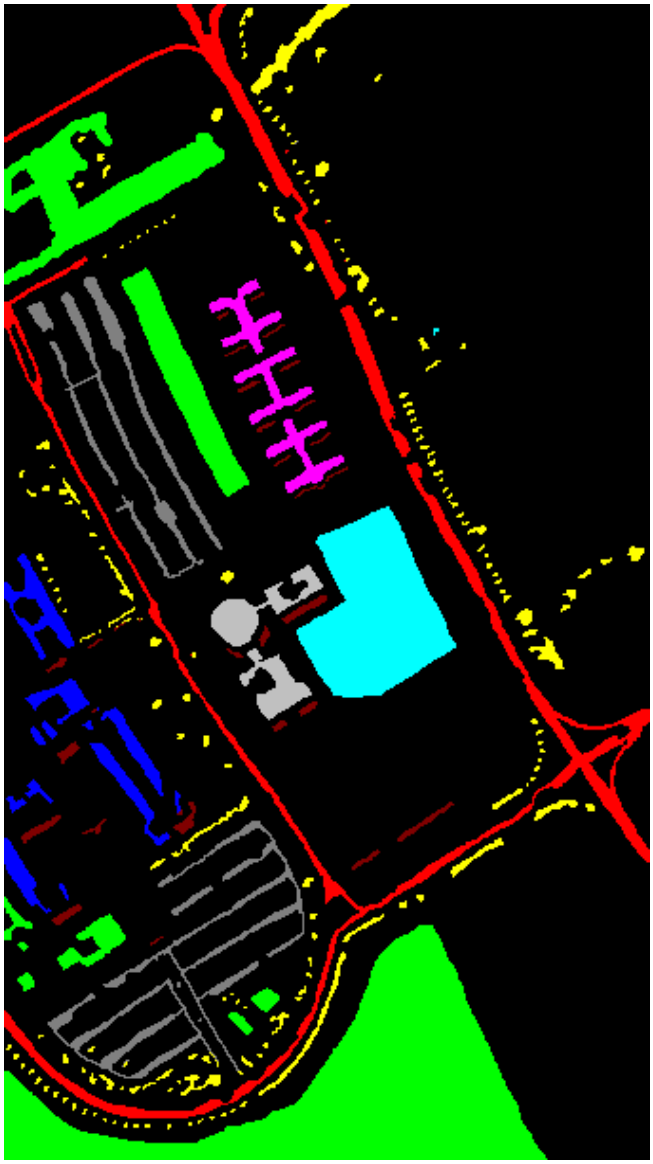


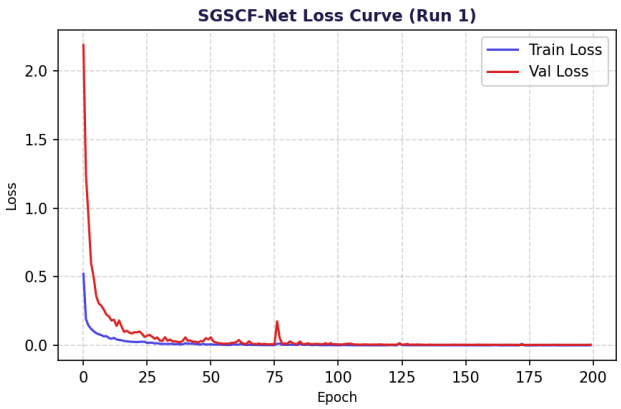
Patch Size 13x13 | Batch Size 128

Overall Metrics:
- **OA (Overall Accuracy):** 99.84%
- **AA (Average Accuracy):** 99.70%
- **Kappa Coefficient:** 99.79
- **Train Time:** 28.2 min

Class-wise Accuracy:
- Asphalt: 99.89%
- Meadows: 100.00%
- Gravel: 99.88%
- Trees: 99.39%
- Painted metal sheets: 100.00%
- Bare Soil: 99.83%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.49%
- Shadows: 98.81%

Visualizations:





Patch Size 13x13 | Batch Size 256

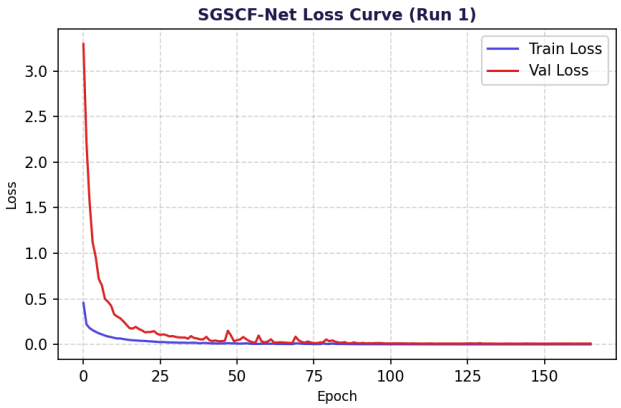
Overall Metrics:
- **OA (Overall Accuracy):** 99.78%
- **AA (Average Accuracy):** 99.55%
- **Kappa Coefficient:** 99.70
- **Train Time:** 24.4 min

Class-wise Accuracy:
- Asphalt: 99.89%
- Meadows: 100.00%
- Gravel: 99.47%
- Trees: 98.95%
- Painted metal sheets: 99.91%
- Bare Soil: 99.83%

- Bitumen: 100.00%
- Self-Blocking Bricks: 99.42%
- Shadows: 98.54%

Visualizations:





Patch Size 15x15 | Batch Size 32

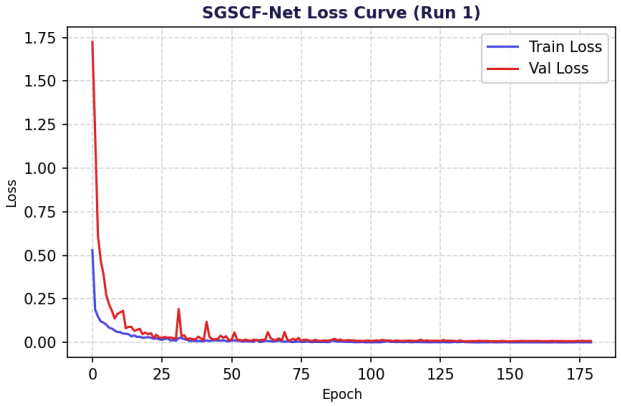
Overall Metrics:
- **OA (Overall Accuracy):** 99.84%
- **AA (Average Accuracy):** 99.65%
- **Kappa Coefficient:** 99.78
- **Train Time:** 38.0 min

Class-wise Accuracy:
- Asphalt: 99.87%
- Meadows: 99.97%

- Gravel: 99.70%
- Trees: 99.35%
- Painted metal sheets: 100.00%
- Bare Soil: 100.00%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.62%
- Shadows: 98.30%

Visualizations:





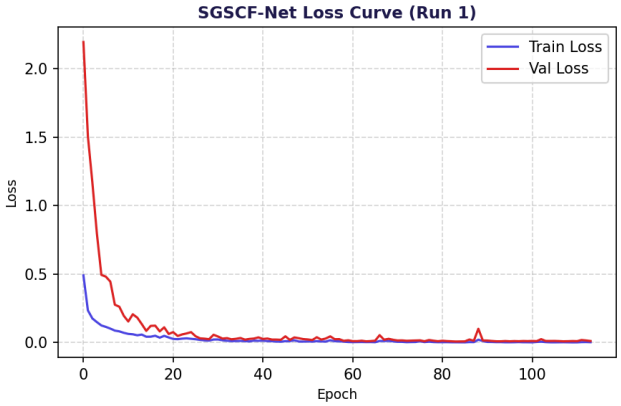
Patch Size 15x15 | Batch Size 64

Overall Metrics:
- **OA (Overall Accuracy):** 99.75%
- **AA (Average Accuracy):** 99.59%
- **Kappa Coefficient:** 99.66
- **Train Time:** 23.6 min

Class-wise Accuracy:
- Asphalt: 99.78%
- Meadows: 99.96%
- Gravel: 99.64%
- Trees: 98.99%
- Painted metal sheets: 99.73%
- Bare Soil: 99.80%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.32%
- Shadows: 99.07%

Visualizations:





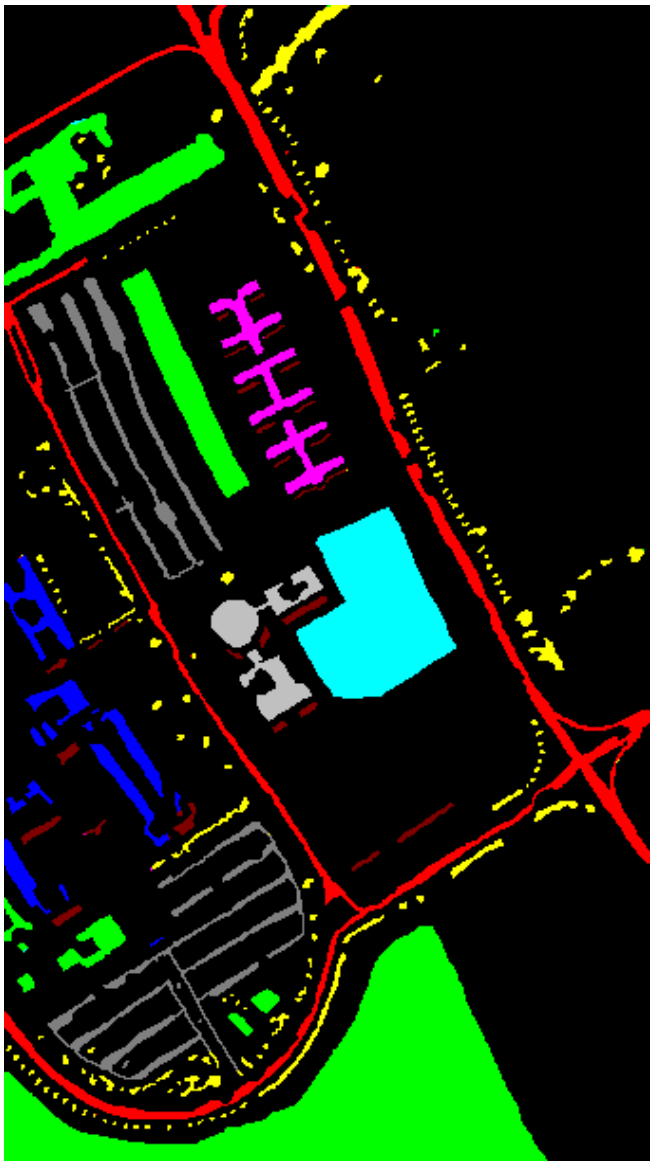
Patch Size 15x15 | Batch Size 128

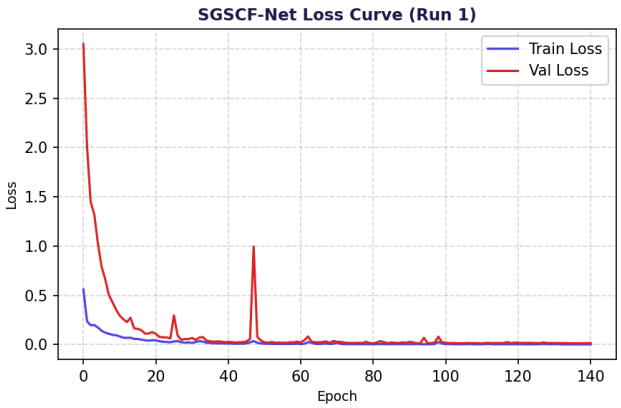
Overall Metrics:
- **OA (Overall Accuracy):** 99.73%

- **AA (Average Accuracy):** 99.53%
- **Kappa Coefficient:** 99.64
- **Train Time:** 29.6 min

Class-wise Accuracy:
- Asphalt: 99.85%
- Meadows: 99.93%
- Gravel: 99.17%
- Trees: 99.07%
- Painted metal sheets: 99.63%
- Bare Soil: 99.75%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.42%
- Shadows: 98.93%

Visualizations:





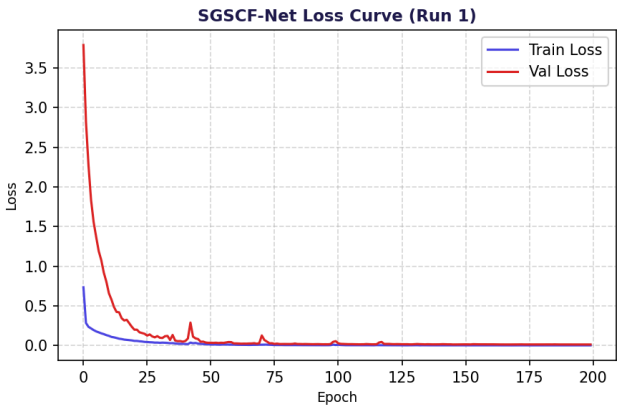
Patch Size 15x15 | Batch Size 256

Overall Metrics:
- **OA (Overall Accuracy):** 99.72%
- **AA (Average Accuracy):** 99.54%
- **Kappa Coefficient:** 99.62
- **Train Time:** 44.4 min

Class-wise Accuracy:
- Asphalt: 99.76%
- Meadows: 99.94%
- Gravel: 99.59%
- Trees: 99.10%
- Painted metal sheets: 100.00%
- Bare Soil: 99.85%
- Bitumen: 99.91%
- Self-Blocking Bricks: 99.01%
- Shadows: 98.68%

Visualizations:





Patch Size 17x17 | Batch Size 32

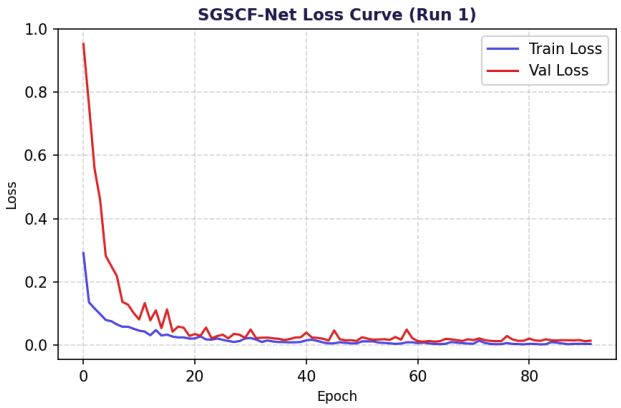
Overall Metrics:
- **OA (Overall Accuracy):** 99.72%
- **AA (Average Accuracy):** 99.31%
- **Kappa Coefficient:** 99.62
- **Train Time:** 24.0 min

Class-wise Accuracy:
- Asphalt: 99.74%
- Meadows: 99.95%
- Gravel: 99.70%
- Trees: 98.94%
- Painted metal sheets: 100.00%
- Bare Soil: 100.00%

- Bitumen: 100.00%
- Self-Blocking Bricks: 99.56%
- Shadows: 95.92%

Visualizations:





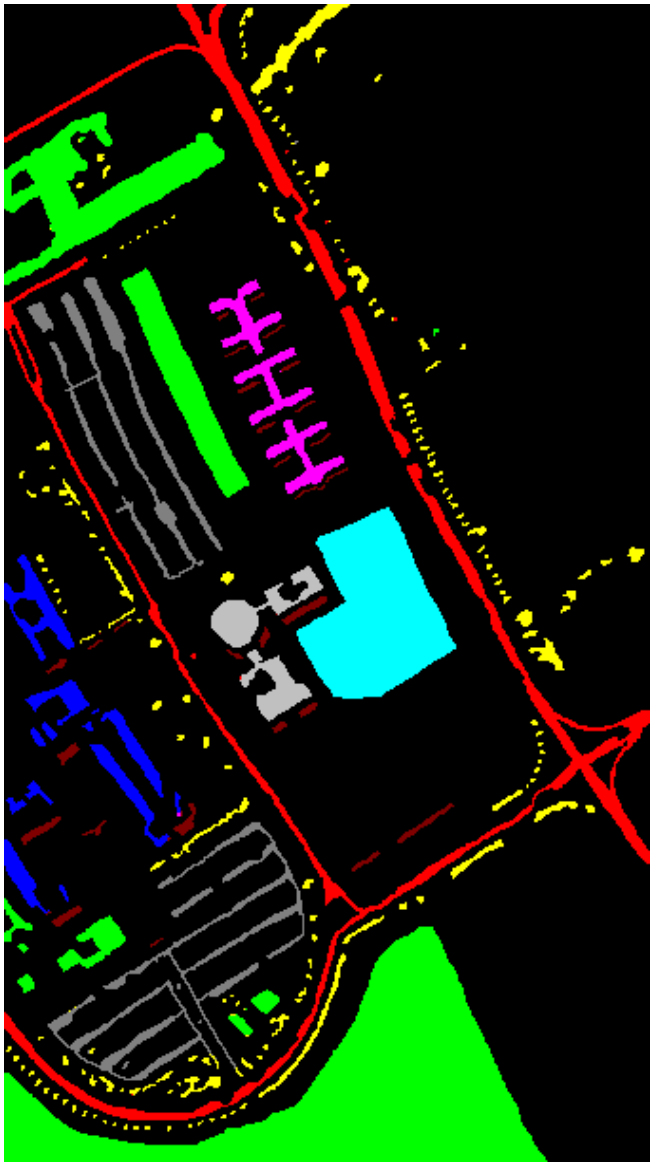
Patch Size 17x17 | Batch Size 64

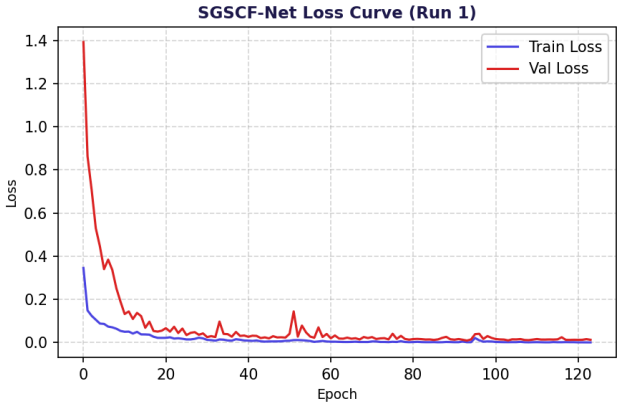
Overall Metrics:
- **OA (Overall Accuracy):** 99.70%
- **AA (Average Accuracy):** 99.50%
- **Kappa Coefficient:** 99.60
- **Train Time:** 32.1 min

Class-wise Accuracy:
- Asphalt: 99.57%
- Meadows: 99.91%

- Gravel: 99.70%
- Trees: 98.98%
- Painted metal sheets: 99.63%
- Bare Soil: 100.00%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.29%
- Shadows: 98.42%

Visualizations:





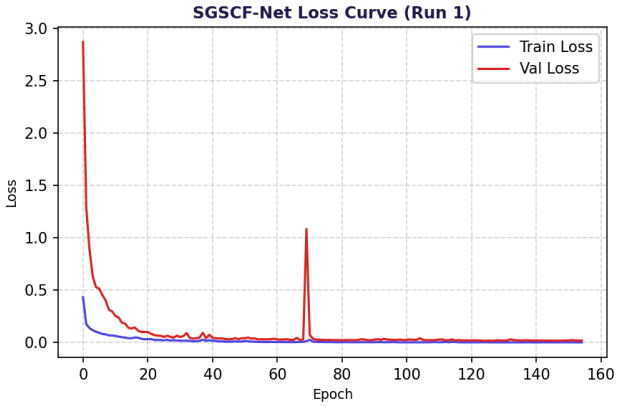
Patch Size 17x17 | Batch Size 128

Overall Metrics:
- **OA (Overall Accuracy):** 99.71%
- **AA (Average Accuracy):** 99.46%
- **Kappa Coefficient:** 99.62
- **Train Time:** 41.2 min

Class-wise Accuracy:
- Asphalt: 99.68%
- Meadows: 99.94%
- Gravel: 99.06%
- Trees: 99.34%
- Painted metal sheets: 99.63%
- Bare Soil: 100.00%
- Bitumen: 100.00%
- Self-Blocking Bricks: 99.22%
- Shadows: 98.26%

Visualizations:





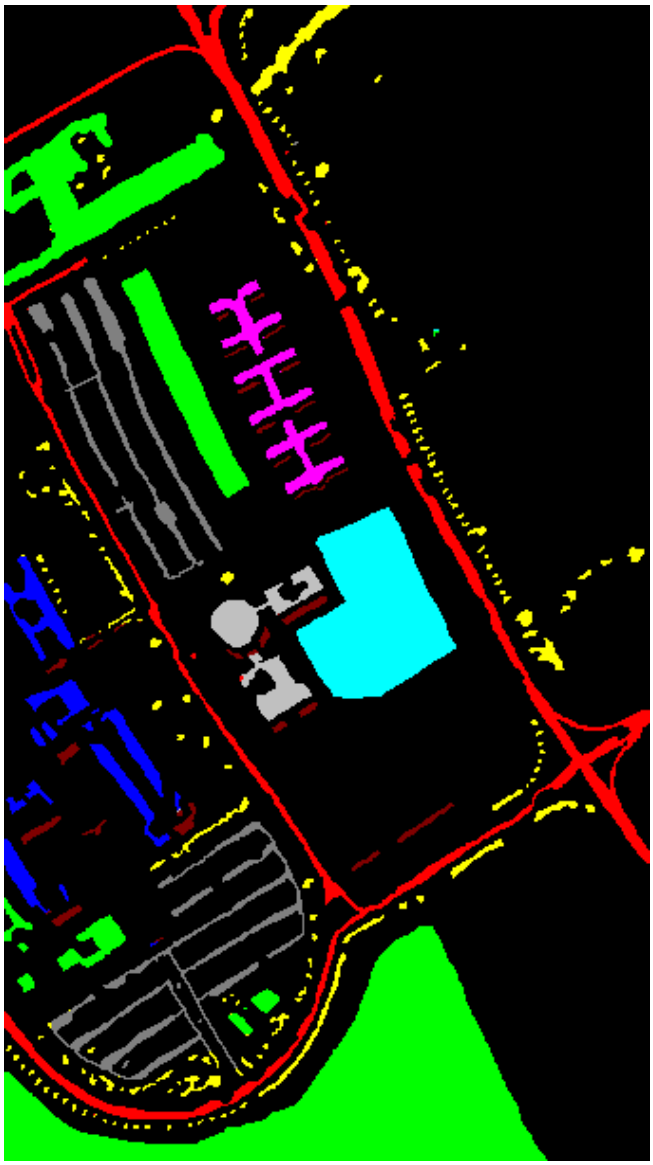
Patch Size 17x17 | Batch Size 256

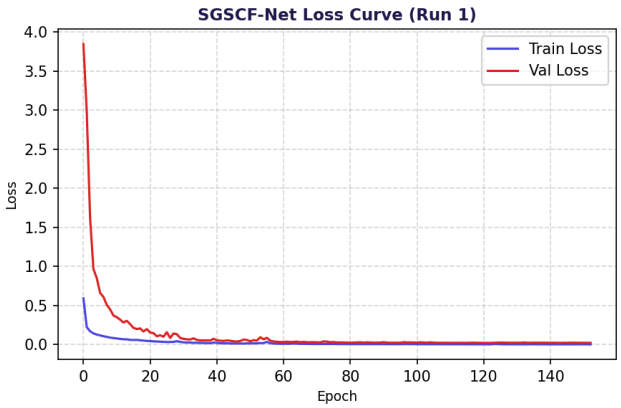
Overall Metrics:
- **OA (Overall Accuracy):** 99.63%

- **AA (Average Accuracy):** 99.19%
- **Kappa Coefficient:** 99.52
- **Train Time:** 38.2 min

Class-wise Accuracy:
- Asphalt: 99.68%
- Meadows: 99.96%
- Gravel: 99.47%
- Trees: 98.54%
- Painted metal sheets: 99.82%
- Bare Soil: 99.93%
- Bitumen: 99.81%
- Self-Blocking Bricks: 99.28%
- Shadows: 96.23%

Visualizations:





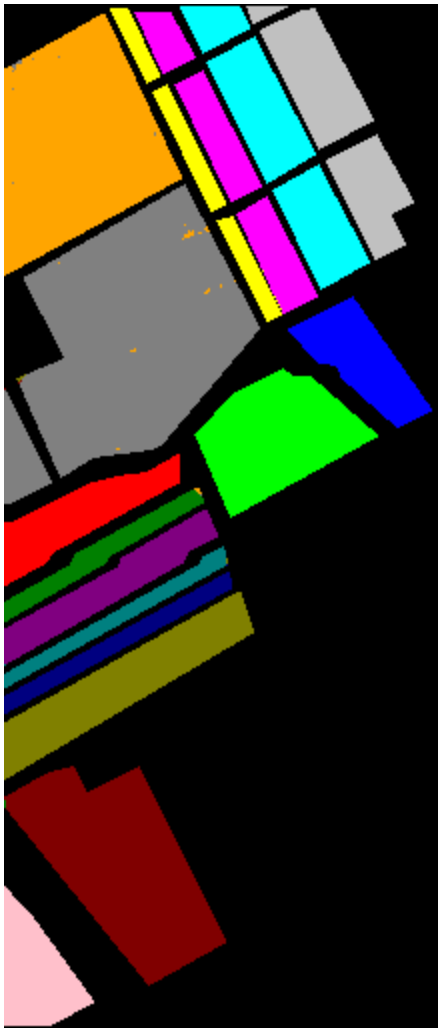
Salinas Valley

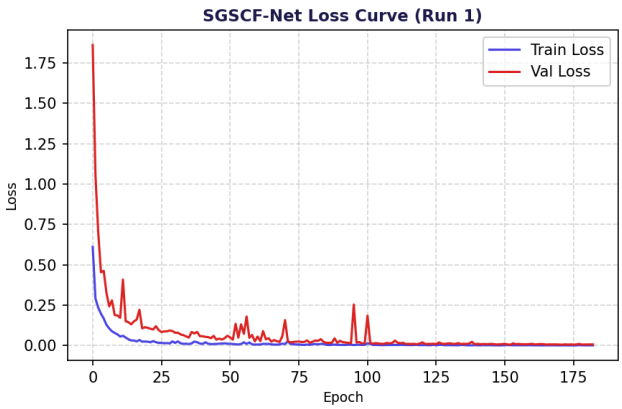
Patch Size 9x9 | Batch Size 32

Overall Metrics:
- **OA (Overall Accuracy):** 99.83%
- **AA (Average Accuracy):** 99.87%
- **Kappa Coefficient:** 99.81
- **Train Time:** 39.8 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.38%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 99.84%
- Soil_vinyard_develop: 99.90%
- Corn_senesced_green_weeds: 99.43%
- Lettuce_romaine_4wk: 99.88%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 99.88%
- Vinyard_untrained: 98.86%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





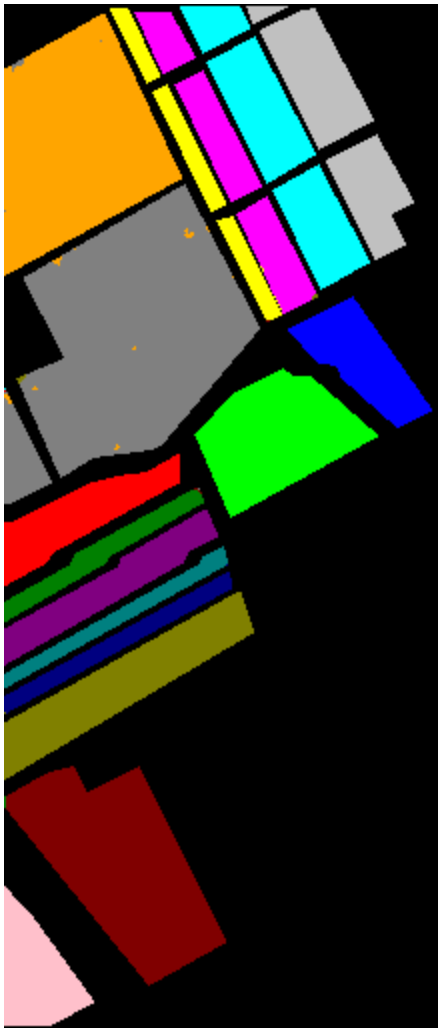
Patch Size 9x9 | Batch Size 64

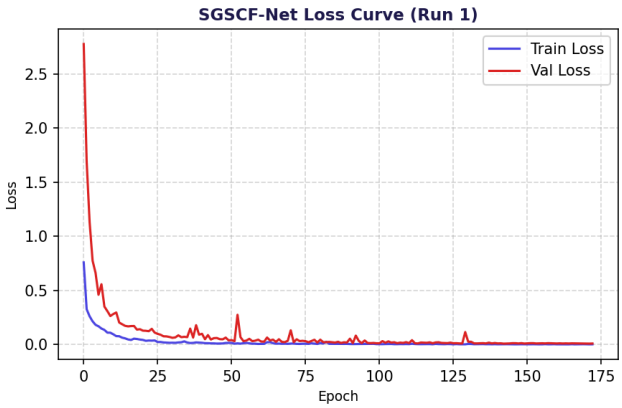
Overall Metrics:
- **OA (Overall Accuracy):** 99.76%
- **AA (Average Accuracy):** 99.81%
- **Kappa Coefficient:** 99.74
- **Train Time:** 40.4 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.64%
- Fallow_smooth: 100.00%
- Stubble: 99.97%
- Celery: 100.00%
- Grapes_untrained: 99.92%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.43%

- Lettuce_romaine_4wk: 98.96%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.73%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.32%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





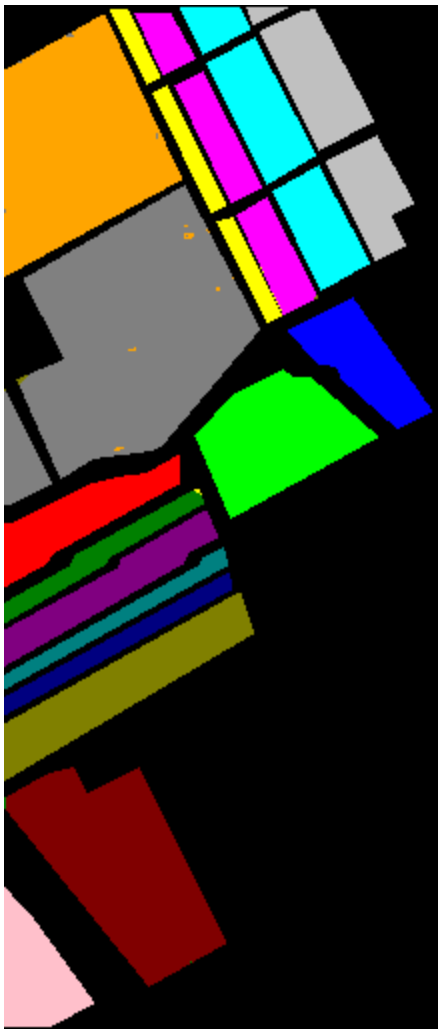
Patch Size 9x9 | Batch Size 128

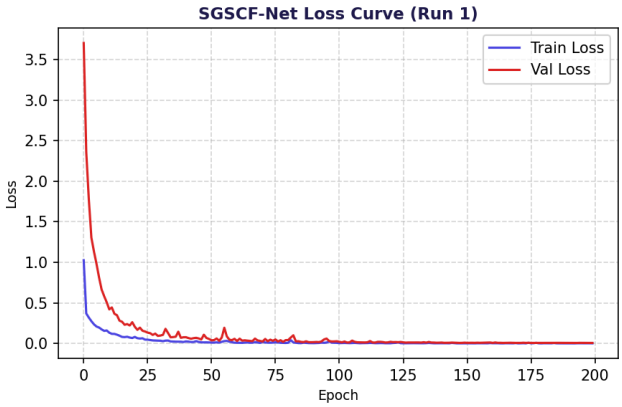
Overall Metrics:
- **OA (Overall Accuracy):** 99.83%
- **AA (Average Accuracy):** 99.82%
- **Kappa Coefficient:** 99.81
- **Train Time:** 49.7 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%

- Fallow_rough_plow: 98.94%
- Fallow_smooth: 99.95%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 99.82%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.47%
- Lettuce_romaine_4wk: 99.53%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.62%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





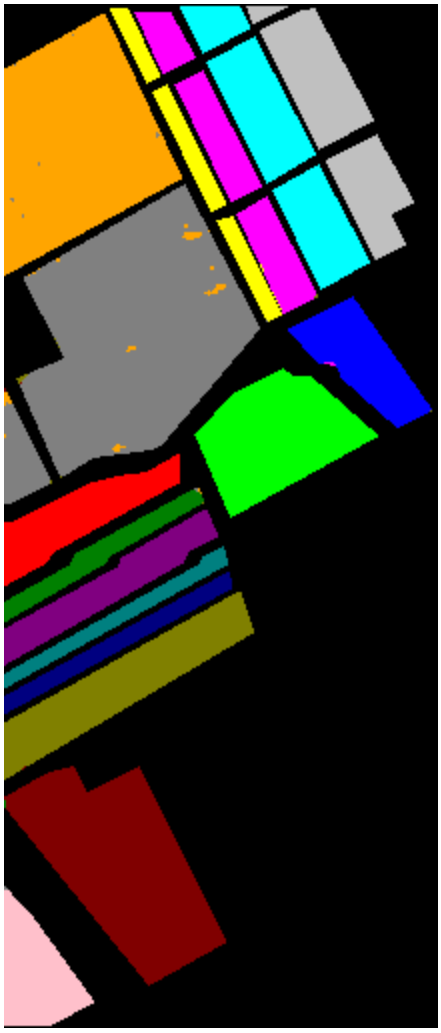
Patch Size 9x9 | Batch Size 256

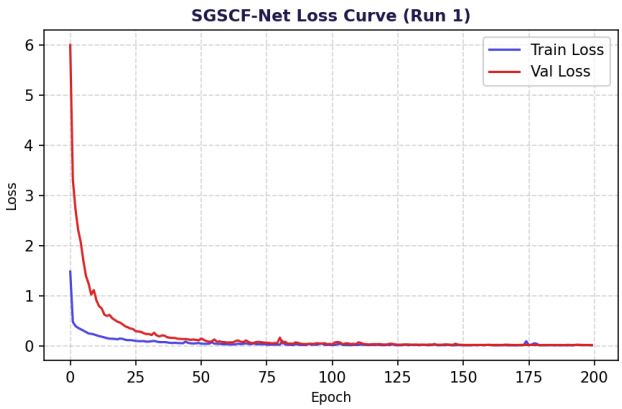
Overall Metrics:
- **OA (Overall Accuracy):** 99.60%
- **AA (Average Accuracy):** 99.73%

- **Kappa Coefficient:** 99.56
- **Train Time:** 54.5 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.56%
- Fallow_smooth: 99.63%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 99.67%
- Soil_vinyard_develop: 99.88%
- Corn_senesced_green_weeds: 99.32%
- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 98.50%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 11x11 | Batch Size 32

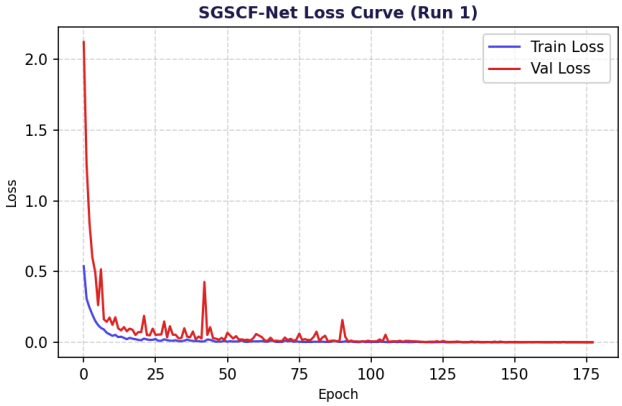
Overall Metrics:
- **OA (Overall Accuracy):** 99.92%
- **AA (Average Accuracy):** 99.92%
- **Kappa Coefficient:** 99.91
- **Train Time:** 57.7 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.64%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 99.93%
- Grapes_untrained: 99.92%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 100.00%

- Lettuce_romaine_4wk: 99.65%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.73%
- Lettuce_romaine_7wk: 99.88%
- Vinyard_untrained: 99.73%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 11x11 | Batch Size 64

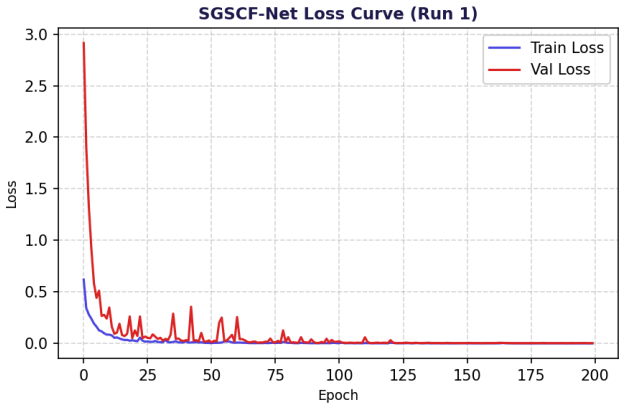
Overall Metrics:
- **OA (Overall Accuracy):** 99.92%
- **AA (Average Accuracy):** 99.92%
- **Kappa Coefficient:** 99.91
- **Train Time:** 69.7 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%

- Fallow_rough_plow: 100.00%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 99.93%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.81%
- Lettuce_romaine_4wk: 99.42%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 99.88%
- Vinyard_untrained: 99.62%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 11x11 | Batch Size 128

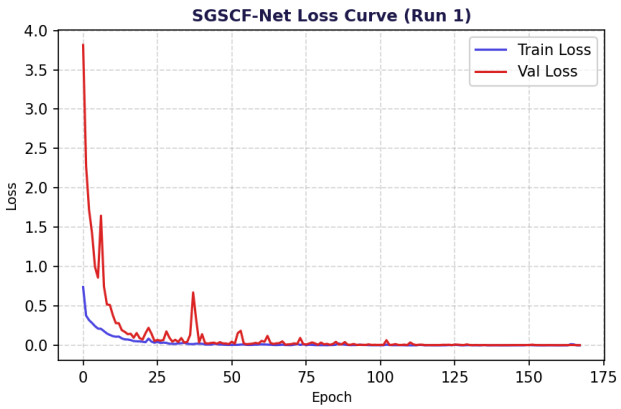
Overall Metrics:
- **OA (Overall Accuracy):** 99.87%
- **AA (Average Accuracy):** 99.89%

- **Kappa Coefficient:** 99.85
- **Train Time:** 65.4 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 99.94%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.47%
- Fallow_smooth: 100.00%
- Stubble: 99.97%
- Celery: 100.00%
- Grapes_untrained: 99.97%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.81%
- Lettuce_romaine_4wk: 99.65%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.55%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 11x11 | Batch Size 256

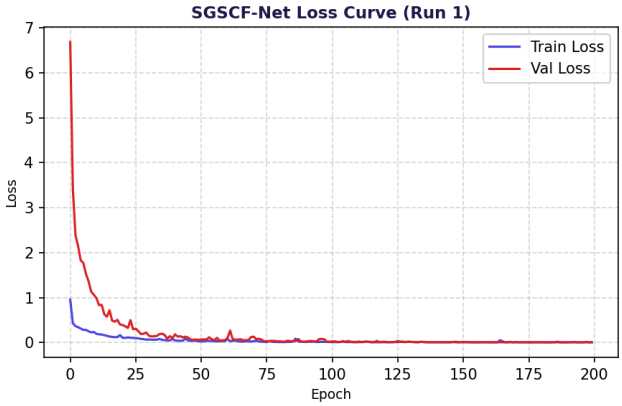
Overall Metrics:
- **OA (Overall Accuracy):** 99.91%
- **AA (Average Accuracy):** 99.93%
- **Kappa Coefficient:** 99.90
- **Train Time:** 79.5 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 99.94%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.73%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 99.97%
- Grapes_untrained: 99.99%
- Soil_vinyard_develop: 99.98%
- Corn_senesced_green_weeds: 99.62%

- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.66%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 13x13 | Batch Size 32

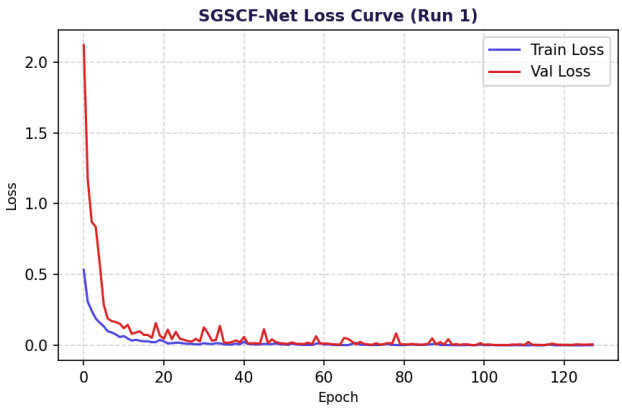
Overall Metrics:
- **OA (Overall Accuracy):** 99.92%
- **AA (Average Accuracy):** 99.92%
- **Kappa Coefficient:** 99.91
- **Train Time:** 43.4 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%

- Fallow_rough_plow: 99.47%
- Fallow_smooth: 100.00%
- Stubble: 99.94%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.92%
- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.61%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 13x13 | Batch Size 64

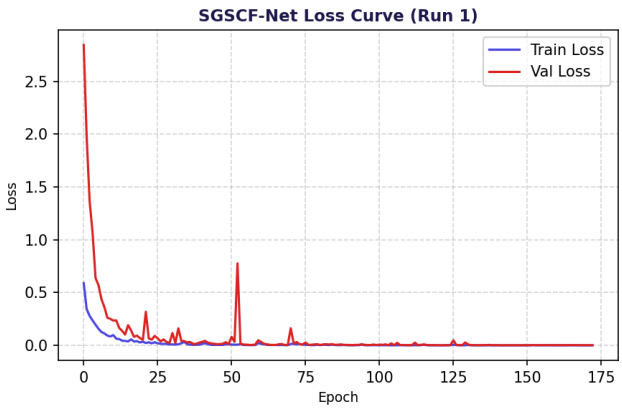
Overall Metrics:
- **OA (Overall Accuracy):** 99.97%
- **AA (Average Accuracy):** 99.96%

- **Kappa Coefficient:** 99.97
- **Train Time:** 57.3 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.82%
- Fallow_smooth: 100.00%
- Stubble: 99.97%
- Celery: 99.97%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.85%
- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.93%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 13x13 | Batch Size 128

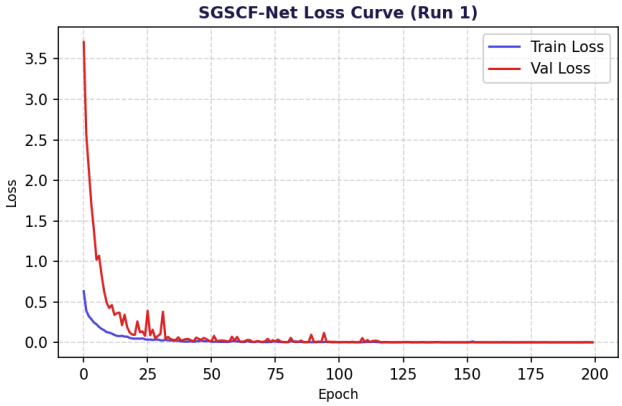
Overall Metrics:
- **OA (Overall Accuracy):** 99.95%
- **AA (Average Accuracy):** 99.93%
- **Kappa Coefficient:** 99.94
- **Train Time:** 67.0 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.38%
- Fallow_smooth: 100.00%
- Stubble: 99.97%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.89%

- Lettuce_romaine_4wk: 99.77%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.85%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 13x13 | Batch Size 256

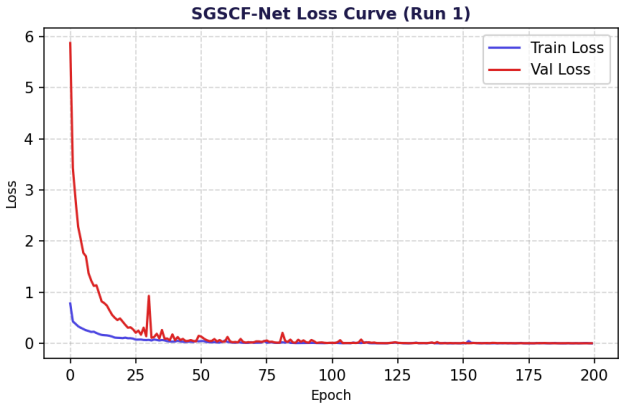
Overall Metrics:
- **OA (Overall Accuracy):** 99.88%
- **AA (Average Accuracy):** 99.86%
- **Kappa Coefficient:** 99.86
- **Train Time:** 59.5 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 99.75%

- Fallow_rough_plow: 98.94%
- Fallow_smooth: 99.95%
- Stubble: 99.97%
- Celery: 100.00%
- Grapes_untrained: 99.98%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.70%
- Lettuce_romaine_4wk: 99.88%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.59%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 15x15 | Batch Size 32

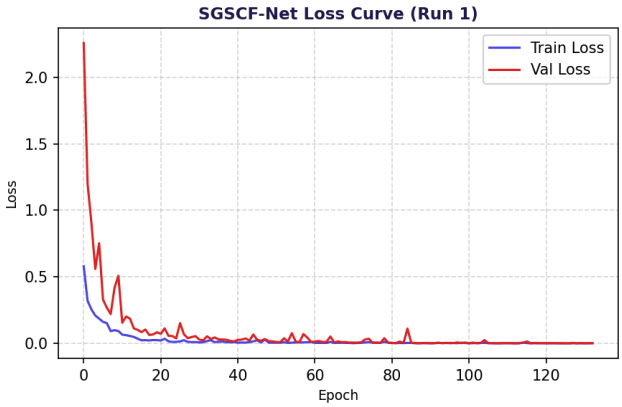
Overall Metrics:
- **OA (Overall Accuracy):** 99.97%
- **AA (Average Accuracy):** 99.94%

- **Kappa Coefficient:** 99.96
- **Train Time:** 63.5 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 99.94%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.64%
- Fallow_smooth: 99.86%
- Stubble: 99.97%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.96%
- Lettuce_romaine_4wk: 99.77%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.97%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 15x15 | Batch Size 64

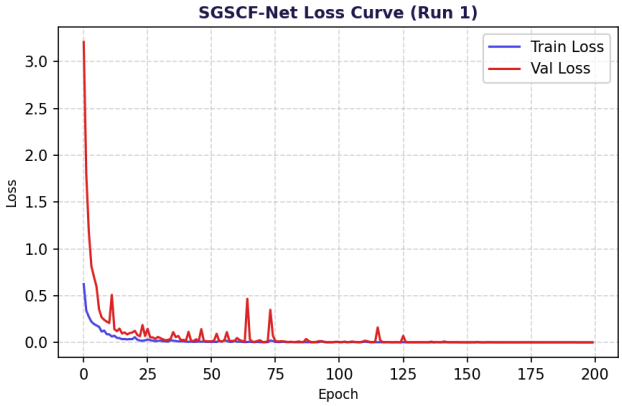
Overall Metrics:
- **OA (Overall Accuracy):** 99.98%
- **AA (Average Accuracy):** 99.97%
- **Kappa Coefficient:** 99.98
- **Train Time:** 93.9 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.91%
- Fallow_smooth: 99.95%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.81%

- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 100.00%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 15x15 | Batch Size 128

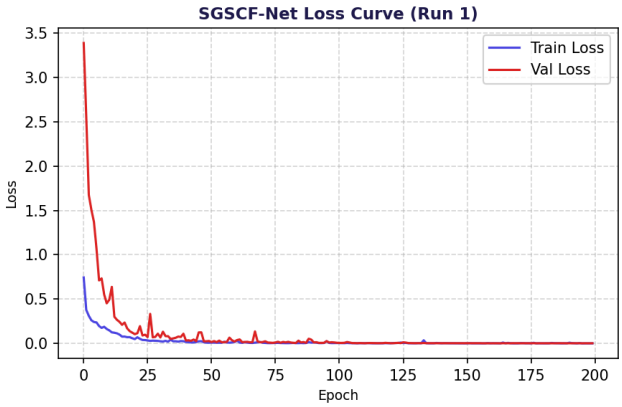
Overall Metrics:
- **OA (Overall Accuracy):** 99.96%
- **AA (Average Accuracy):** 99.93%
- **Kappa Coefficient:** 99.96
- **Train Time:** 95.1 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%

- Fallow_rough_plow: 99.64%
- Fallow_smooth: 99.95%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.73%
- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.59%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.97%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 15x15 | Batch Size 256

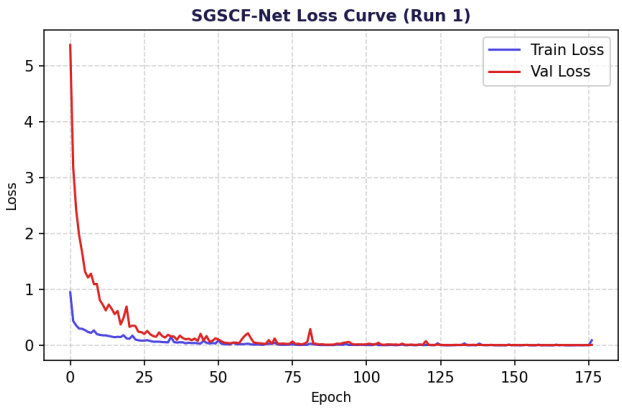
Overall Metrics:
- **OA (Overall Accuracy):** 99.92%
- **AA (Average Accuracy):** 99.82%

- **Kappa Coefficient:** 99.91
- **Train Time:** 74.6 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.73%
- Fallow_smooth: 99.91%
- Stubble: 100.00%
- Celery: 99.97%
- Grapes_untrained: 99.99%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.62%
- Lettuce_romaine_4wk: 98.96%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 98.92%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.98%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 17x17 | Batch Size 32

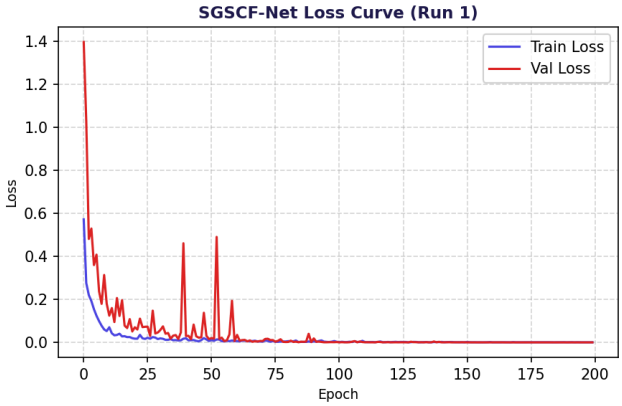
Overall Metrics:
- **OA (Overall Accuracy):** 100.00%
- **AA (Average Accuracy):** 100.00%
- **Kappa Coefficient:** 100.00
- **Train Time:** 116.8 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 100.00%
- Fallow_smooth: 99.95%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 100.00%

- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 100.00%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 17x17 | Batch Size 64

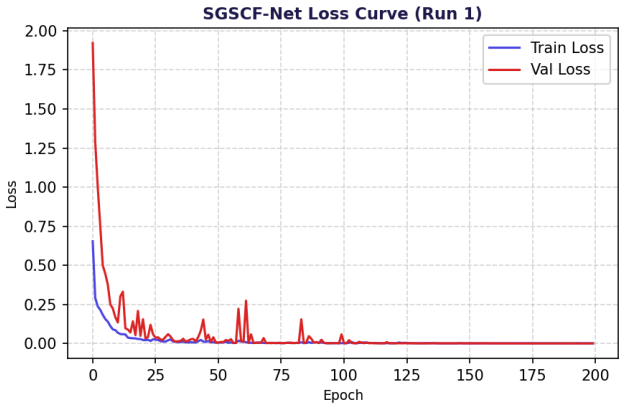
Overall Metrics:
- **OA (Overall Accuracy):** 100.00%
- **AA (Average Accuracy):** 99.99%
- **Kappa Coefficient:** 100.00
- **Train Time:** 115.4 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%

- Fallow_rough_plow: 99.91%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 100.00%
- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 100.00%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 17x17 | Batch Size 128

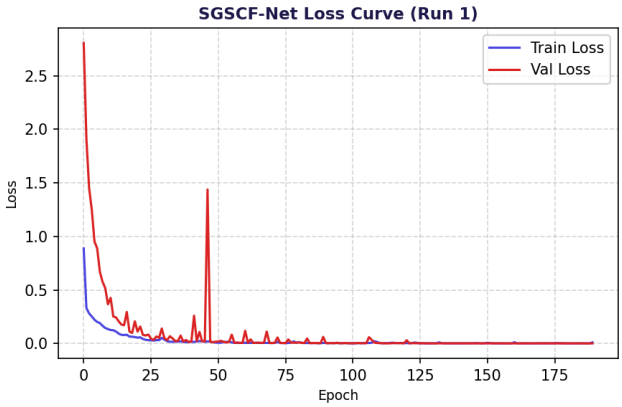
Overall Metrics:
- **OA (Overall Accuracy):** 99.97%
- **AA (Average Accuracy):** 99.94%

- **Kappa Coefficient:** 99.97
- **Train Time:** 94.2 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.64%
- Fallow_smooth: 99.95%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 100.00%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 100.00%
- Lettuce_romaine_4wk: 99.65%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.97%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Patch Size 17x17 | Batch Size 256

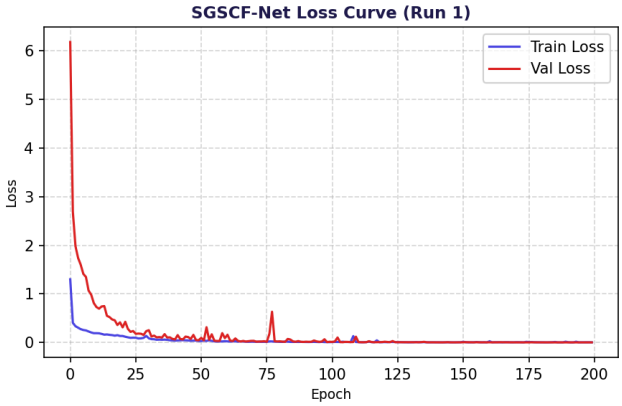
Overall Metrics:
- **OA (Overall Accuracy):** 99.97%
- **AA (Average Accuracy):** 99.94%
- **Kappa Coefficient:** 99.97
- **Train Time:** 111.0 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.12%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 99.99%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 100.00%

- Lettuce_romaine_4wk: 100.00%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 99.86%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 100.00%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





PART 2: Train Ratio Experiments (Data Scarcity)

Indian Pines (Train Ratio Experiment)

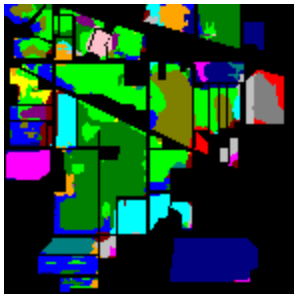
Training Ratio: 1%

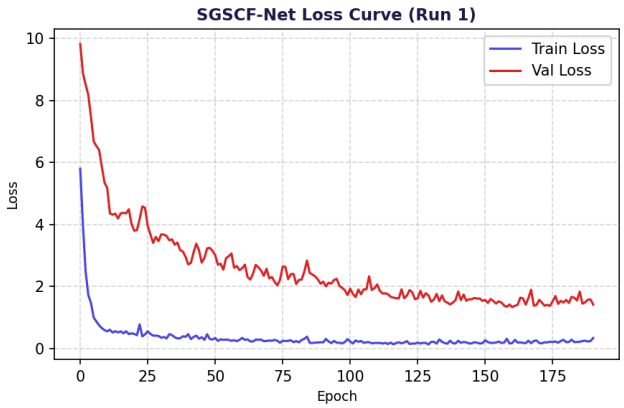
Overall Metrics:
- **OA (Overall Accuracy):** 70.12%
- **AA (Average Accuracy):** 66.23%

- **Kappa Coefficient:** 66.07
- **Train Time:** 1.2 min

Class-wise Accuracy:
- Alfalfa: 25.00%
- Corn-notill: 52.37%
- Corn-mintill: 52.90%
- Corn: 92.11%
- Grass-pasture: 86.49%
- Grass-trees: 84.18%
- Grass-pasture-mowed: 8.66%
- Hay-windrowed: 99.27%
- Oats: 7.66%
- Soybean-notill: 73.27%
- Soybean-mintill: 87.10%
- Soybean-clean: 94.89%
- Wheat: 71.84%
- Woods: 86.45%
- Buildings-Grass-Trees: 63.87%
- Stone-Steel-Towers: 73.68%

Visualizations:





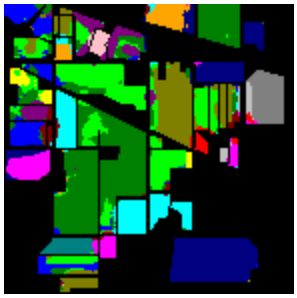
Training Ratio: 2%

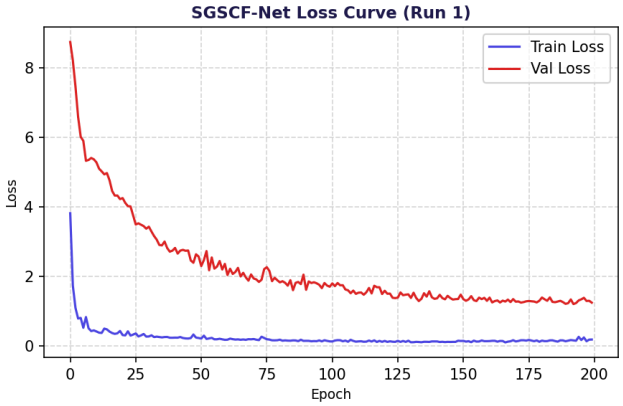
Overall Metrics:
- **OA (Overall Accuracy):** 81.55%
- **AA (Average Accuracy):** 76.57%
- **Kappa Coefficient:** 78.86
- **Train Time:** 2.4 min

Class-wise Accuracy:
- Alfalfa: 52.44%
- Corn-notill: 67.29%
- Corn-mintill: 87.23%
- Corn: 80.51%
- Grass-pasture: 91.56%
- Grass-trees: 91.05%

- Grass-pasture-mowed: 26.51%
- Hay-windrowed: 99.75%
- Oats: 10.94%
- Soybean-notill: 79.98%
- Soybean-mintill: 82.19%
- Soybean-clean: 86.34%
- Wheat: 96.74%
- Woods: 87.36%
- Buildings-Grass-Trees: 96.64%
- Stone-Steel-Towers: 88.64%

Visualizations:





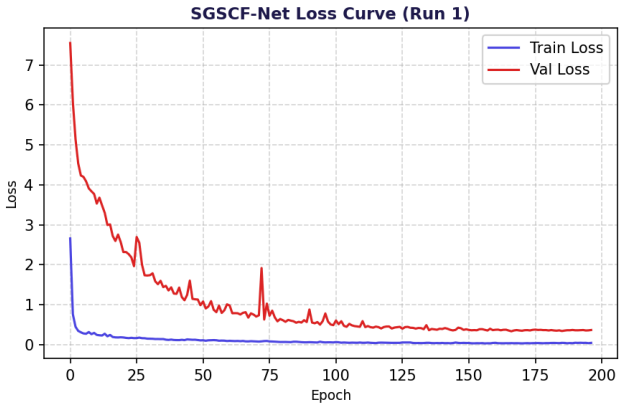
Training Ratio: 5%

Overall Metrics:
- **OA (Overall Accuracy):** 93.91%
- **AA (Average Accuracy):** 86.65%
- **Kappa Coefficient:** 93.06
- **Train Time:** 5.8 min

Class-wise Accuracy:
- Alfalfa: 73.68%
- Corn-notill: 94.41%
- Corn-mintill: 91.75%
- Corn: 97.03%
- Grass-pasture: 98.23%
- Grass-trees: 96.31%
- Grass-pasture-mowed: 47.06%
- Hay-windrowed: 96.75%
- Oats: 42.86%
- Soybean-notill: 87.79%
- Soybean-mintill: 95.74%
- Soybean-clean: 93.33%
- Wheat: 97.27%
- Woods: 95.57%
- Buildings-Grass-Trees: 95.17%
- Stone-Steel-Towers: 83.51%

Visualizations:





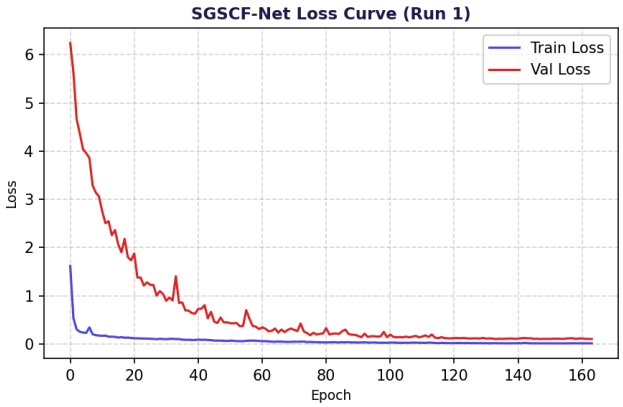
Training Ratio: 10%

Overall Metrics:
- **OA (Overall Accuracy):** 97.84%
- **AA (Average Accuracy):** 93.43%
- **Kappa Coefficient:** 97.55
- **Train Time:** 9.5 min

Class-wise Accuracy:
- Alfalfa: 92.50%
- Corn-notill: 98.10%
- Corn-mintill: 96.97%
- Corn: 100.00%
- Grass-pasture: 99.22%
- Grass-trees: 98.84%
- Grass-pasture-mowed: 90.91%
- Hay-windrowed: 100.00%
- Oats: 45.16%
- Soybean-notill: 98.47%
- Soybean-mintill: 98.42%
- Soybean-clean: 94.26%
- Wheat: 99.39%
- Woods: 98.27%
- Buildings-Grass-Trees: 99.01%
- Stone-Steel-Towers: 85.37%

Visualizations:





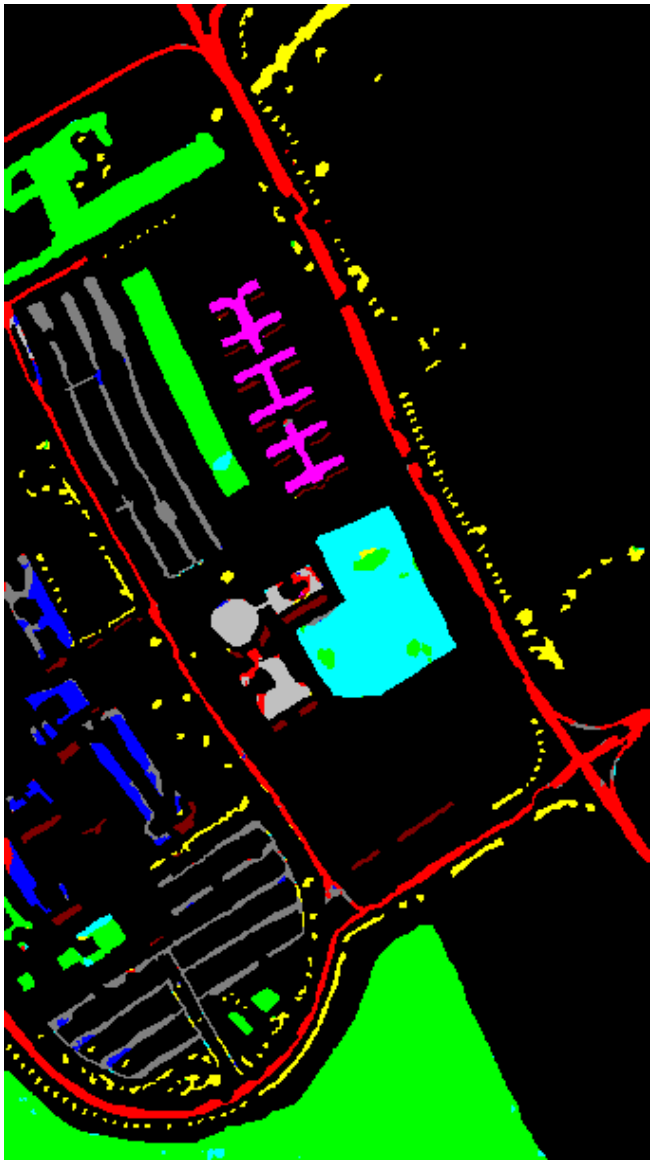
Pavia University (Train Ratio Experiment)

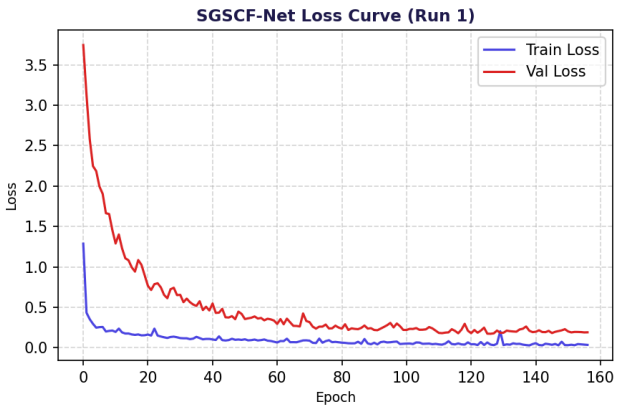
Training Ratio: 1%

Overall Metrics:
- **OA (Overall Accuracy):** 95.41%
- **AA (Average Accuracy):** 94.48%
- **Kappa Coefficient:** 93.90
- **Train Time:** 2.1 min

Class-wise Accuracy:
- Asphalt: 95.54%
- Meadows: 98.13%
- Gravel: 89.42%
- Trees: 95.14%
- Painted metal sheets: 99.62%
- Bare Soil: 94.27%
- Bitumen: 95.80%
- Self-Blocking Bricks: 84.70%
- Shadows: 97.66%

Visualizations:





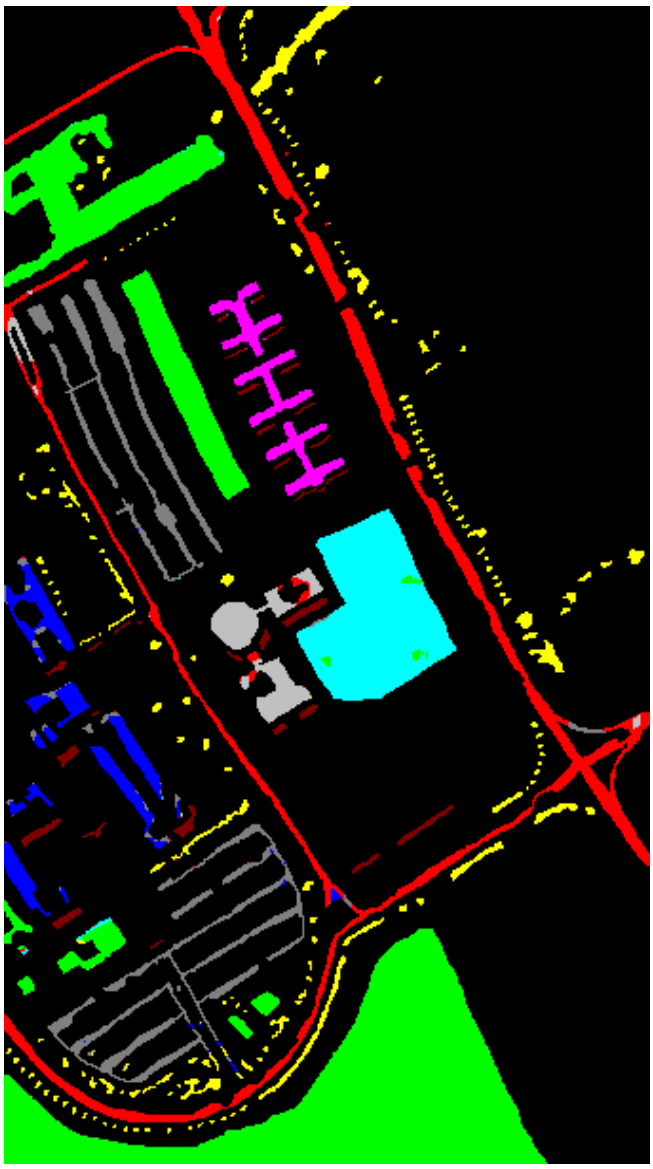
Training Ratio: 2%

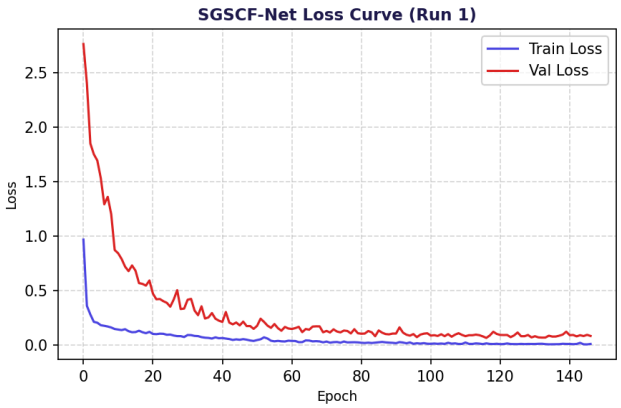
Overall Metrics:
- **OA (Overall Accuracy):** 98.04%
- **AA (Average Accuracy):** 96.85%
- **Kappa Coefficient:** 97.40
- **Train Time:** 3.9 min

Class-wise Accuracy:
- Asphalt: 98.04%
- Meadows: 99.53%
- Gravel: 96.83%
- Trees: 98.68%
- Painted metal sheets: 99.54%
- Bare Soil: 98.98%

- Bitumen: 89.08%
- Self-Blocking Bricks: 92.32%
- Shadows: 98.69%

Visualizations:





Training Ratio: 5%

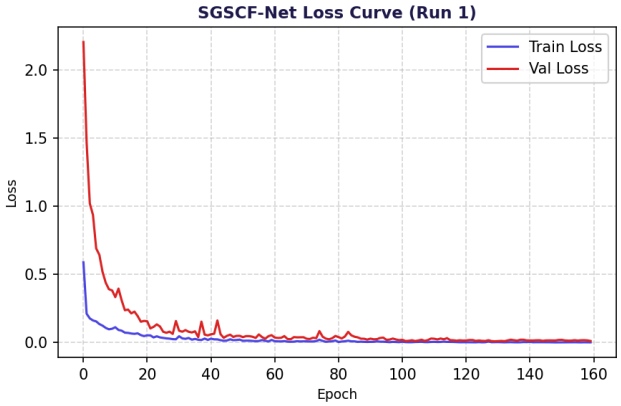
Overall Metrics:
- **OA (Overall Accuracy):** 99.51%
- **AA (Average Accuracy):** 99.09%
- **Kappa Coefficient:** 99.35
- **Train Time:** 10.7 min

Class-wise Accuracy:
- Asphalt: 99.50%
- Meadows: 99.98%

- Gravel: 99.08%
- Trees: 99.13%
- Painted metal sheets: 99.92%
- Bare Soil: 99.91%
- Bitumen: 97.46%
- Self-Blocking Bricks: 97.90%
- Shadows: 98.96%

Visualizations:





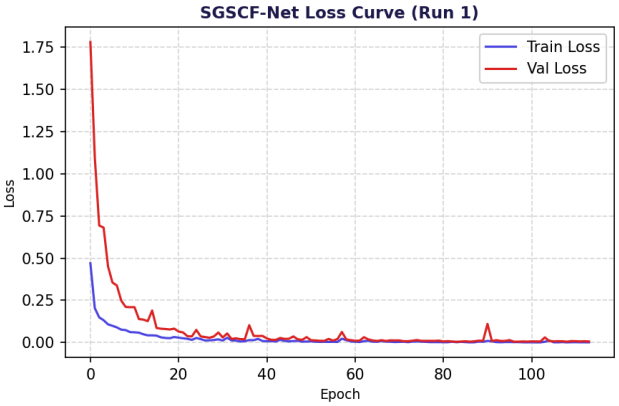
Training Ratio: 10%

Overall Metrics:
- **OA (Overall Accuracy):** 99.83%
- **AA (Average Accuracy):** 99.73%
- **Kappa Coefficient:** 99.78
- **Train Time:** 15.2 min

Class-wise Accuracy:
- Asphalt: 99.79%
- Meadows: 100.00%
- Gravel: 99.82%
- Trees: 99.35%
- Painted metal sheets: 100.00%
- Bare Soil: 99.93%
- Bitumen: 99.72%
- Self-Blocking Bricks: 99.39%
- Shadows: 99.61%

Visualizations:





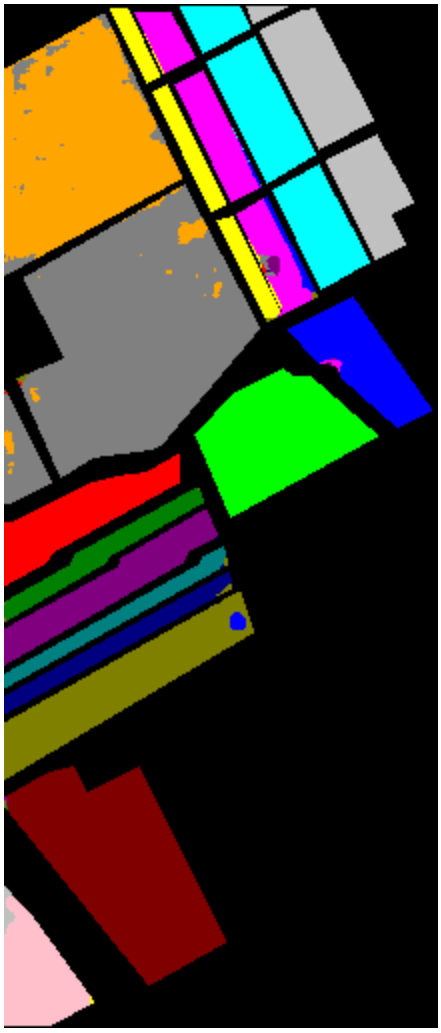
Salinas Valley (Train Ratio Experiment)

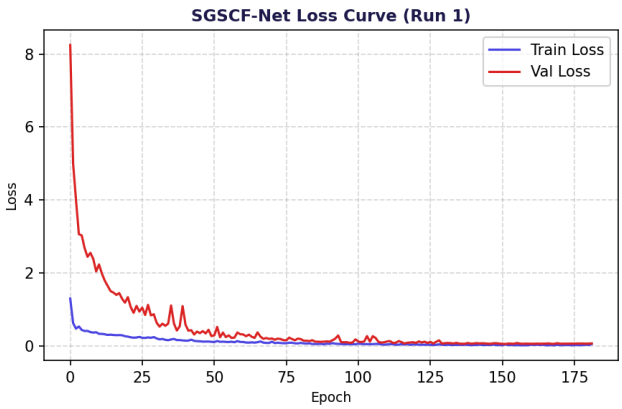
Training Ratio: 1%

Overall Metrics:
- **OA (Overall Accuracy):** 97.73%
- **AA (Average Accuracy):** 98.03%
- **Kappa Coefficient:** 97.48
- **Train Time:** 5.7 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 99.39%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 89.47%
- Fallow_rough_plow: 96.31%
- Fallow_smooth: 98.85%
- Stubble: 99.87%
- Celery: 98.26%
- Grapes_untrained: 96.29%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 98.07%
- Lettuce_romaine_4wk: 99.81%
- Lettuce_romaine_5wk: 96.81%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 99.23%
- Vinyard_untrained: 96.10%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Training Ratio: 2%

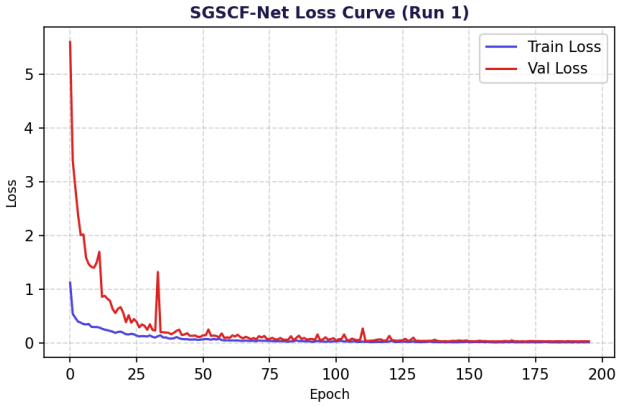
Overall Metrics:
- **OA (Overall Accuracy):** 99.11%
- **AA (Average Accuracy):** 99.19%
- **Kappa Coefficient:** 99.00
- **Train Time:** 12.4 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 99.97%
- Fallow: 99.89%
- Fallow_rough_plow: 97.59%
- Fallow_smooth: 100.00%
- Stubble: 99.82%
- Celery: 100.00%
- Grapes_untrained: 98.62%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 98.27%

- Lettuce_romaine_4wk: 98.18%
- Lettuce_romaine_5wk: 99.57%
- Lettuce_romaine_6wk: 98.31%
- Lettuce_romaine_7wk: 99.70%
- Vinyard_untrained: 97.67%
- Vinyard_vertical_trellis: 99.48%

Visualizations:





Training Ratio: 5%

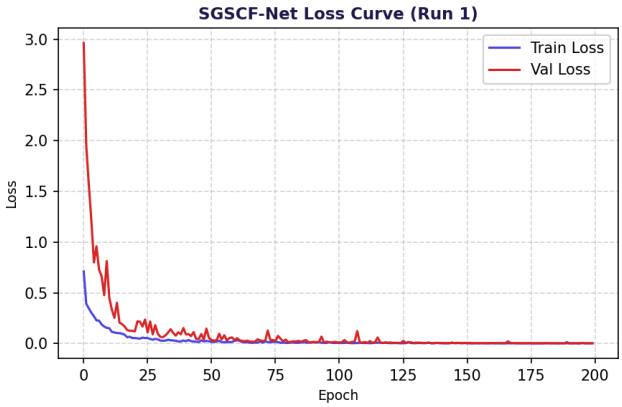
Overall Metrics:
- **OA (Overall Accuracy):** 99.79%
- **AA (Average Accuracy):** 99.77%
- **Kappa Coefficient:** 99.76
- **Train Time:** 31.5 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 99.89%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%

- Fallow_rough_plow: 99.77%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 99.70%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.49%
- Lettuce_romaine_4wk: 98.67%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 99.38%
- Vinyard_untrained: 99.50%
- Vinyard_vertical_trellis: 100.00%

Visualizations:





Training Ratio: 10%

Overall Metrics:
- **OA (Overall Accuracy):** 99.93%
- **AA (Average Accuracy):** 99.93%

- **Kappa Coefficient:** 99.92
- **Train Time:** 57.8 min

Class-wise Accuracy:
- Brocoli_green_weeds_1: 100.00%
- Brocoli_green_weeds_2: 100.00%
- Fallow: 100.00%
- Fallow_rough_plow: 99.38%
- Fallow_smooth: 100.00%
- Stubble: 100.00%
- Celery: 100.00%
- Grapes_untrained: 99.94%
- Soil_vinyard_develop: 100.00%
- Corn_senesced_green_weeds: 99.96%
- Lettuce_romaine_4wk: 99.88%
- Lettuce_romaine_5wk: 100.00%
- Lettuce_romaine_6wk: 100.00%
- Lettuce_romaine_7wk: 100.00%
- Vinyard_untrained: 99.71%
- Vinyard_vertical_trellis: 100.00%

Visualizations:



